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Review

A Review of Methodologies for Photovoltaic Energy Generation Forecasting in the Building Sector

Omid Pedram [1], Ana Soares [1,2] and Pedro Moura [1,3,*]

[1] Department of Electrical and Computer Engineering, University of Coimbra, 3030-290 Coimbra, Portugal; omid.pedram@student.uc.pt (O.P.); ana.soares@inescc.pt (A.S.)
[2] Institute for Systems Engineering and Computers at Coimbra, DEEC, University of Coimbra, 3030-290 Coimbra, Portugal
[3] Institute of Systems and Robotics, DEEC, University of Coimbra, 3030-290 Coimbra, Portugal
* Correspondence: pmoura@isr.uc.pt

Abstract

Photovoltaic (PV) systems are swiftly expanding within the building sector, offering significant benefits such as renewable energy integration, yet introducing challenges due to mismatches between local generation and demand. With the increasing availability of data and advanced modeling tools, stakeholders are increasingly motivated to adopt energy management and optimization techniques, where accurate forecasting of PV generation is essential. While the existing literature provides valuable insights, a comprehensive review of methodologies specifically tailored for the forecast of PV generation in buildings remains scarce. This study aims to address this gap by analyzing the forecasting methods, data requirements, and performance metrics employed, with the primary objective of providing an in-depth review of previous research. The findings highlight the critical role of improving PV energy generation forecasting accuracy in enhancing energy management and optimization for individual buildings. Additionally, the study identifies key challenges and opportunities for future research, such as the limited exploration of localized environmental and operational factors (such as partial shading, dust, and dirt); insufficient data on building-specific PV output patterns; and the need to account for variability in PV generation. By clarifying the current state of PV energy forecasting methodologies, this research lays essential groundwork for future advancements in the field.

Keywords: building-attached photovoltaic; building-integrated photovoltaic; photovoltaic power generation forecasting; Artificial Intelligence; Artificial Neural Networks; Machine Learning

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1. Introduction

1.1. Motivation

About 40% of primary energy [1] and 60% of the world's electricity are consumed by residential and commercial buildings, contributing to about 33% of global greenhouse gas emissions [2]. Hence, the building sector has become a significant concern for policymakers and scholars, with the drive for green technologies focused on the minimization of environmental impacts [3] and energy costs [4]. Within this framework, photovoltaic (PV) systems stand out as the preferred choice to be integrated into buildings [5], primarily due to their capacity to produce decentralized electricity [6] and seamless integration into architectural structures [7], using the building-attached PV (BAPV) and building-integrated

PV (BIPV) options. Furthermore, technological advancements and market implementation have driven down the cost of PV panels, rendering them increasingly appealing for integration into building infrastructure [8].

Building on this momentum, the adoption of PV systems in buildings is rapidly increasing. For instance, the building-integrated photovoltaics (BIPV) market is experiencing rapid growth, with a projected compound annual growth rate of 21% between 2024 and 2030 [9]. Also, global rooftop solar capacity has continued its rapid growth, increasing from 118 GW in 2022 [10] to 159 GW in 2023 [11], with projections reaching nearly 190 GW by the end of 2024 [12].

However, the integration of solar energy into buildings presents a challenge due to its inherent variability [13]. Solar power is available only during daylight hours and is susceptible to the impact of climate, geographical location, seasonal variations, and time of day. Additionally, the performance of PV modules may be impacted by the buildup of dust in the summer [14] and snowfall during the winter [15]. Hence, the growing integration of variable and intermittent renewable energy sources can lead to a misalignment between energy supply and demand [16], potentially disrupting power system stability, efficiency, quality, and reliability [17].

To address the mismatch between energy supply and demand, a range of solutions can be found in the existing body of literature. Incorporating buildings into communities [18] by facilitating the sharing of surplus renewable energy generation stands as a viable solution for addressing this challenge. Additionally, Energy Storage Systems (ESSs) and Demand Response (DR) programs assume vital roles in bestowing the requisite flexibility to ensure the matching between renewable generation and energy demand [19]. In this framework, flexible energy resources assume an essential role within DR initiatives, enabling the minimization of peak demand and maximization of the matching with renewable generation availability [20]. In addition, electric vehicles play a crucial role in promoting energy flexibility by serving as DR (with charging management) or ESS (with vehicle-to-building) resources [21].

In the context of the mismatch between renewable energy generation and demand, accurate forecasting in both cases emerges as a crucial tool for effectively managing energy flexibility resources [22] and consumption in buildings [23]. Moreover, it facilitates the efficient utilization of ESSs by providing insights into periods with potential renewable generation surplus or deficit [24].

The forecasting algorithms are typically integrated within Building Energy Management Systems (BEMSs) [25]. BEMSs gather and process data from diverse sources, including weather stations; historical energy consumption records; grid information [26]; and the internet of things devices, such as smart meters, occupancy sensors, and temperature monitors, which supply real-time data on energy consumption patterns and building conditions [27]. Subsequently, the algorithms analyze this data to forecast PV energy generation and energy demand [28]. These forecasts are typically presented as time series data, indicating energy production or consumption at specific intervals, such as hourly or daily [29].

BEMSs commonly use PV generation forecasts as inputs to optimization and scheduling routines that control flexible loads and battery storage. Forecasts are fed into receding horizon controllers or optimization engines to decide when to charge/discharge batteries, shift electric water heating or HVAC preconditioning, and manage EV charging, thereby enabling anticipatory, cost-, and emissions-aware operations.

For example, Amabile et al. [30] studied a residential home equipped with PV and an electric water heater. In this case, the EMS utilized a 30-min PV generation forecast to schedule hot water heating. The forecast-driven optimization boosted PV self-consumption

by about 11% compared to a rule-based controller. Angenendt et al. [31] implement forecast-based operation strategies (tested with perfect and persistence forecasts) combined with variable PV feed-in limits in a residential PV–battery EMS. In simulation, this approach substantially increased the total energy throughput and lifetime of the battery and reduced the cost by approximately 12%. In a university microgrid, Duran et al. [32] incorporated LSTM-based PV power forecasts into a mixed-integer algorithm. By anticipating PV generation (to store energy and dispatch it by EMS when solar resources are scarce) and load forecasts, the EMS manages the state of charge of the battery system. It reduced peak loads, resulting in lower emissions. In their case, the forecast-based strategy reduced CO_2 emissions by 3.58% and covered 59% of the demand.

These examples illustrate that integrating PV forecasts into BEMSs enables anticipatory, optimization-driven control of storage and flexible loads—shifting consumption into high-PV periods, reducing curtailment and peak demand, and moderating battery cycling—which yields higher self-consumption, lower operating costs and emissions, and reduced grid stress. Advanced management strategies, supported by forecasting models, optimize renewable integration, besides contributing to more efficient energy usage. Ultimately, these advancements have the potential to make buildings more sustainable and responsive to energy demands.

Therefore, given the challenges posed by energy demand and renewable energy's variability within the building sector, accurate forecasting of energy production and demand is essential. However, to realize these benefits in practice, forecasts must explicitly account for building-specific factors (partial shading, soiling, inverter topology, and façade orientation) that increase uncertainty in on-site PV output.

1.2. Research Gap and Contributions

According to the review articles, numerous studies have explored the forecasting of demand in buildings [33–36] and the prediction of PV output [37–41]. The authors of [37,38] aimed to highlight the significance of ML and the importance of data preprocessing and parameter selection in predicting solar energy production. In [39], Erdener et al. concentrated on PV prediction applications for aggregate, regional behind-the-meter production effective to bulk systems and utility operations. Sobri et al. [40] examined a range of methods for solar production forecasting, including time series statistical methods, physical methods, and ensemble methods. In 2019, Viscondi et al. conducted a systematic review to explore the role of big data in solar energy production forecasting [41].

To our best understanding, there is currently no review article exclusively dedicated to forecasting PV generation, specifically within the building sector. The novelty of this paper lies in providing a systematic, in-depth review of PV generation forecasting methods specific to the building sector. The contributions of this paper are fourfold: (1) It identifies the diverse objectives and constraints associated with building-scale PV forecasting, (2) it analyzes the various techniques and computational methods applied in recent studies, (3) it examines the different metrics utilized in recent researches, and (4) it highlights areas where the current methodologies fall short, particularly in integrating emerging technologies and sustainable practices. Ultimately, this work provides a comprehensive perspective on the state-of-the-art methods in building-scale PV forecasting, offering critical insights that can inform future research and guide innovation in this crucial area.

1.3. Paper Organization

The subsequent sections of this paper are structured as follows. Section 2 outlines the literature review methodology, detailing the keywords, databases, and encountered limitations. Section 3 examines PV generation forecasting in buildings, emphasizing

building-specific peculiarities affecting PV performance, data quality, key datasets, forecasting methods, real-world use cases, and emerging forecasting methods. It also explores PV adoption across building types, associated challenges, and evaluation approaches, covering metrics, post-processing, and uncertainty quantification. Additionally, a review of studies since 2019 highlights diverse forecasting approaches and algorithms. Section 4 presents key findings, discusses the literature limitations, and proposes a research framework. Finally, Section 5 summarizes the study's main contributions and implications.

2. Literature Review Methodology

This section details the approach used to identify and analyze the relevant research on PV energy generation forecasting in the building sector. It outlines the search strategy, selection criteria, and limitations inherent to the review process.

This literature review used a manual qualitative content analysis approach adapted from Neuendorf [42]. Each included paper was read in full, and key information (forecasting methods, evaluation metrics, geographic region, etc.) was summarized in the body of the article and tables. An inductive thematic synthesis of these extractions was then performed to identify recurring methodological patterns, application gaps, and directions for future research. A comprehensive search was conducted across the Scopus database, using a set of targeted keywords: TITLE-ABS-KEY (photovoltaic AND generation OR production AND forecast* AND building) AND PUBYEAR > 2018 AND PUBYEAR < 2025 AND (LIMIT-TO (SUBJAREA, "ENGI")) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp")) AND (LIMIT-TO (LANGUAGE, "English")). Relevant articles were identified through a thorough review of article titles, abstracts, and contents. The selected articles were then subjected to a rigorous analysis to identify existing solutions, research gaps, and potential advancements to address the identified gaps.

Around 83.15% of the papers on this topic appeared in journals, and 75.28% of all journal and conference papers focused on BAPV.

This study's literature selection was restricted to articles published between 2019 and 2024 to ensure a focus on the most recent advancements in the field. This time frame was chosen due to significant developments in AI techniques for energy forecasting, enabling a current and comprehensive overview of best practices. The choice of keywords has influenced the scope of the review, and thus, future studies could expand keyword selection to include emerging trends and topics. Additionally, this paper offers a narrative review of relevant literature, recognizing that this approach differs from a systematic review. The criteria for including or excluding articles retrieved through the search are outlined in Figure 1.

Furthermore, in this review, the main differences compared to forecasting for power plants are discussed, even though the methods and technologies of PV generation forecasting in the building sector and power plants have many similarities. Both types of forecasting rely on the same fundamental principles, such as the use of historical data [43], weather forecasts [44], and physical models [45] to estimate future energy generation. Additionally, both types of forecasting can benefit from the use of AI techniques, including Artificial Neural Networks (ANNs) and Machine Learning (ML) [46]. They also exhibit distinct characteristics due to their scale (size/capacity and distributed/centralized) [47], data availability, and optimization goals.

PV power forecasting can be categorized based on how far into the future it forecasts, with each horizon useful for different applications. Table 1 describes different forecasting horizons. Rooftop PV forecasting often focuses on a finer granularity, such as hourly or sub-hourly predictions, to align with the dynamic nature of residential energy consumption [48].

Utility-scale forecasting may employ daily or even longer-term predictions to manage grid stability and resource planning [49].

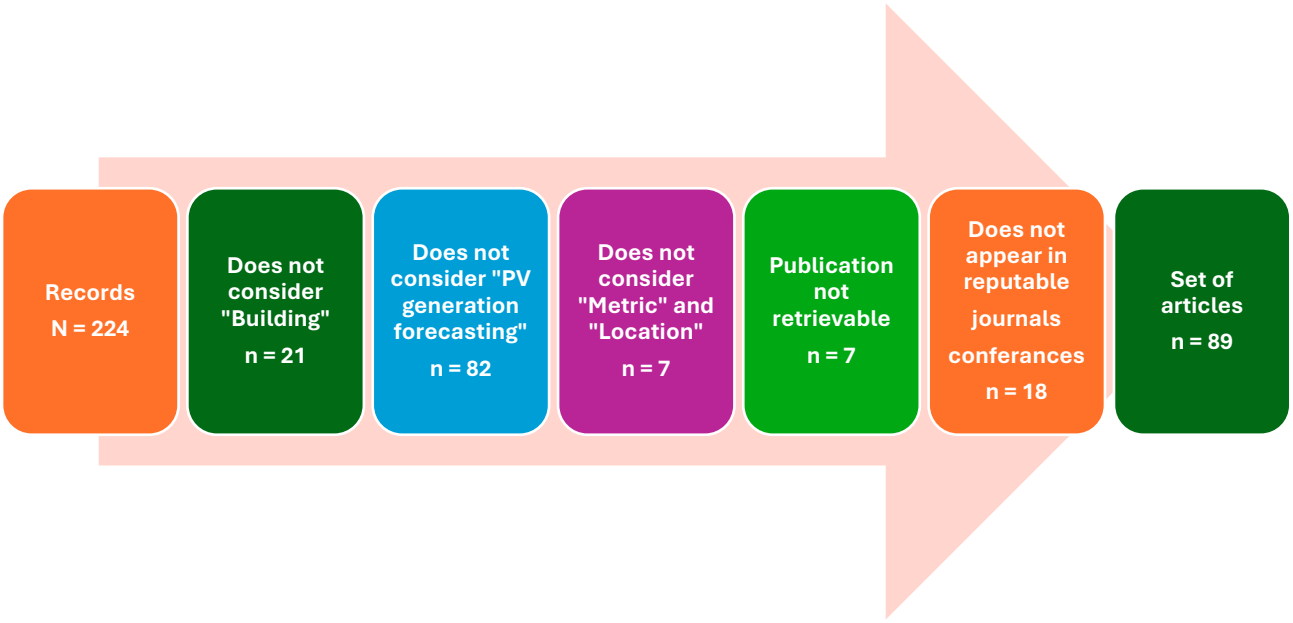


Figure 1. Method of selecting articles.

Table 1. Different forecasting horizons [50].

Classification	Range	Application
Very short-term	Seconds to minutes	Control and management of PV systems, electricity market operations, and microgrid control
Short-term	Up to 72 h	Control of power system operations, unit commitment, and economic dispatch
Medium-term	A few days to a week	Planning maintenance and operations of solar power plants
Long-term	Months to a year	Planning maintenance and operations of solar power plants

Rooftop PV forecasting may have access to more detailed data from individual installations, including shading patterns [51], panel orientation, and historical energy production data [52]. Additionally, power plants have more advanced monitoring systems and can collect more data than rooftop PV arrays, which allows for more accurate forecasting [53]. Finally, rooftop PV forecasting may prioritize optimizing energy generation for individual households or communities, considering aspects like self-consumption, self-sufficiency, and peak shaving [54].

It is important to acknowledge a limitation in the scope of this review, as it exclusively concentrates on the building sector. While this choice enables a detailed examination of rooftop PV forecasting methodologies tailored to residential and small-scale installations, it may not encompass the full spectrum of forecasting challenges and solutions in larger utility-scale power plants. Consequently, this review may not fully represent the unique considerations, methodologies, and technologies specific to utility-scale PV forecasting. Furthermore, the results of this study may not be generalizable to other sectors, which may have different forecasting challenges and solutions.

3. Results

Adequate energy management in buildings is imperative for energy acquisition cost reduction, minimizing energy consumption, and mitigating environmental impact, all while ensuring that user comfort remains uncompromised. As mentioned earlier, forecasting stands out as a powerful tool for achieving these goals. Exploring the field of PV generation forecasting in the building sector, Section 3 provides a comprehensive literature review to categorize and understand the various efforts in forecasting PV generation in buildings. Section 3.1 explains how physical, environmental, and technological factors influence photovoltaic system generation. Section 3.2 highlights the critical role of data quality in ensuring accurate PV generation prediction. Section 3.3 details the different prediction methods. In Section 3.4, there is an anticipation of the evaluation approach's role in addressing challenges within the building sector. Section 3.5 introduces the various building types that utilize PV systems. Section 3.6 offers a thorough review of studies for PV generation forecasting in the building sector, with a specific focus on the diverse approaches and algorithms used in this area of research. Section 3.7 presents real-world applications of PV generation forecasting in buildings. Section 3.8 explores recent approaches to time series forecasting. Finally, the practical implications of forecasting accuracy are explained in Section 3.9.

3.1. Building-Specific Peculiarities/Environment Affecting PV Performance

Building-related characteristics/environment affecting PV performance are classified into two broad groups, as outlined in the sections below.

3.1.1. Physical and Environmental Factors

PV installations in the building sector have unique characteristics that affect generation forecasting. façade-mounted PV experiences different irradiance incidences (a wider range of incidence angles); much more frequent and complex partial shading in urban settings [55,56]; and different thermal boundary conditions (ventilation, wind exposure, and convective heat transfer) [57] that change module back sheet temperatures and therefore electrical output and reliability.

Vertical PV refers to PV modules installed with a near-vertical plane rather than the conventional tilted roof orientation. In general, vertically mounted mono-facial modules produce substantially less annual energy per unit area than optimally tilted arrays under the same location because of higher incidence angles and lower direct irradiance; documents report that optimally tilted arrays can produce 50–70% more annual energy than vertical façades in many cases [58].

However, two important caveats apply. First, the use of vertical bifacial PV can help recover the yield gap, because it harvests reflected and diffuse radiation [59]. In addition, vertical bifacial PV may outperform nearby vertical mono-facial systems under high-albedo, high-latitude, or snow-enhanced conditions [60]. Second, vertical configurations (especially east–west orientations) shift production towards morning and evening hours, which often improves temporal matching with building loads and increases self-consumption values despite lower midday output [60]. Because of these dependencies, accurate yield assessment for vertical PV requires explicit modeling of bifacial gains, ground albedo, view factors, shading, and angle-of-incidence losses.

Partial shading stands as one of the most important causes of performance loss in PV arrays, potentially leading to reduced power output and cell damage if undetected [61]. This issue is particularly critical in the building sector. As noted by IEA-PVPS (2023) [62], BIPVs often impose shading or suboptimal orientation (panels may receive non-ideal sun angles) on PV modules, exacerbating energy losses. Standard forecasting models often

overlook these effects, but they can significantly bias PV output. For example, Kappler et al. [63] developed a Machine Learning shading correction for PV forecasts that uses internal inverter signals and previous day irradiance to detect shading intensity and correct predictions, reporting RMSE reductions of up to 40% on several 10 kWp systems, showing that explicitly modeling shading-related fluctuations substantially improves short-term forecast accuracy.

Similarly, soiling constitutes one of the most critical environmental factors that can be the largest sources of error for rooftop PV forecasts, and incorporating them yields more robust predictions [64]. Li et al. [65] found that, even in a rainy, subtropical city, residual soiling (especially bird droppings) caused up to 15.5% PV power loss. Such nonuniform soiling introduces extra uncertainty into forecasts.

Urban canyons are defined as street-scale spaces bounded by building façades [66], which materially alter the solar resource available to building envelopes and, hence, affect PV forecasting and EMS decisions. Urban canyons are commonly characterized by the height-to-width ratio (H/W) [67] and the sky view factor (SVF). H/W describes the canyon's relative depth and strongly influences canyon shadowing, along with actual shading [67], while SVF (fraction of visible sky) quantifies the sky exposure (range 0–1 means completely obstructed sky to completely open sky) [68], so SVF often correlates with irradiance on façades. In addition, vegetation, canyon orientation, height variability, and building density further modulate both direct and diffuse irradiance [68]. Altogether, all the mentioned factors determine the availability, composition (direct vs. diffuse), and timing of solar irradiance reaching façades and rooftops and, thus, the electrical yield of façade-mounted PV. Considering SVF and vegetation, Rostami et al. [69] found that narrow, tree-lined canyons can reduce incident solar radiation by up to 75%, whereas broad canyons ($H/W < 0.5$) received roughly four times the radiation of narrow ($H/W > 1.8$), shaded canyons. The same study reported that 65% of total solar radiation was received by rooftops, highlighting the potential of rooftops to install PV systems.

As a result, PV generation is influenced by physical and environmental factors. These include the building façade, vertical PV installations, partial shading, soiling, and urban canyon effects, all of which can significantly alter the amount of solar energy captured and the overall system performance.

3.1.2. Technology Factors

Inverter topology describes the electronic architecture that converts PV DC to grid-compatible AC and implements system-level controls such as Maximum Power Point Tracking (MPPT) [70]. Topologies range from centralized/string inverters (many modules series-connected to a single DC–AC stage) to hybrid systems with per module DC optimizers to fully distributed Module-Level Power Electronics (MLPEs) such as microinverters that perform per module MPPT and DC–AC conversion [71]. Figure 2 shows a simplified schematic of the different types of inverters.

String inverters remain the dominant, cost-efficient choice. However, they require uniform module orientation/tilt within each string and are sensitive to partial shading or individual panel faults; when a panel is bypassed, the performance of the whole string can decline [71]. Therefore, Module-Level Power Electronics (MLPEs) devices were applied to address these challenges. Microinverters are commonly used MLPE devices that mitigate these issues by providing per module MPPT, reducing mismatch and partial shading losses, and improving yield under nonuniform irradiance. Microinverters also support module-level rapid shutdown for safety, greater design flexibility, and detailed monitoring and remote diagnostics at a higher cost [71].

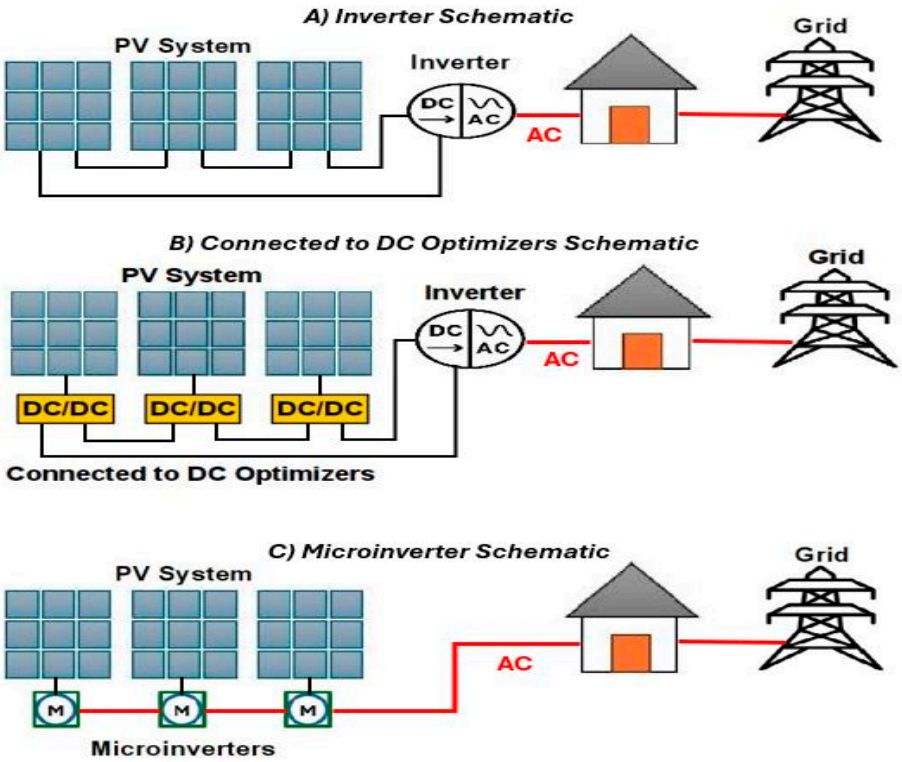


Figure 2. Different types of inverters schematic.

Different inverter architectures provide different telemetry profiles; microinverters typically expose module-level signals, while centralized/string inverters usually report string- or inverter-level aggregates. Inverters offer rich module-level telemetry, which is highly useful for building energy management and reproducible forecasting studies. Modern systems typically expose instantaneous AC power and cumulative energy, device temperature, MPPT status, grid voltage/frequency, power factor, and fault/status codes [72]. These signals are aggregated by a local gateway and exported via PLC or IP to cloud APIs [72]. Module-level data enable fault localization, façade-level performance attribution, and tighter integration with BEMSs for control evaluation.

Consequently, the inverter topology class (central/string, optimizer, and microinverter) influences the model complexity, error metrics, and the interpretation of performance differences between studies.

3.2. The Role of High-Quality Data in PV Generation Forecasting in the Building Sector

Data play an important role in forecasting PV generation in buildings, where accurate forecasts are essential for effective energy management, increasing self-sufficiency and self-consumption, and reducing electricity bills.

Forecasting models rely on historical and real-time data to identify patterns, trends, and anomalies. Accurate input data, such as historical solar radiation and PV generation records, are crucial for capturing the dynamic and intermittent nature of solar energy and PV generation. Additionally, incorporating building-specific factors, such as shading effects, enhances the accuracy of the forecasts. Poor or incomplete data can result in unreliable predictions, adversely affect energy planning, and increase dependency on the grid. Given the critical role of data in the field, Table 2 introduces the relevant databases.

Table 2. Databases.

Data Focusing on	Region/Explanation	References
Weather data	Global	[73–76]
	The USA	[77,78]
	Korea	[79]
	Swedish Meteorological and Hydrological Institute (SMHI)	[80]
	- Desert Knowledge Australia Solar Center (DKASC) - Quality data and knowledge related to solar power	[81]
Solar data	Global	[82–84]
PV generation data	Global	[85,86]
Weather variables and PV generation data	Global	[87,88]
	Global level + Other data sets related to power systems	[89]
	Global level + Other data sets related to energy	[90]
	A case study + Other related data sets	[91,92]
Wind data	Global	[93]

3.3. Types of Forecasting Methods

A comprehensive taxonomy of prediction models encompasses physical, statistical, hybrid, and ensemble methodologies [50]. A brief description of the methods is given below.

3.3.1. Physical Methods

Physical methods for PV forecasting encompass a variety of approaches, including Numerical Weather Prediction (NWP) models and PV module performance models. NWP models [94], which can also be applied to satellite imagery [95] to forecast the state of the weather conditions, operate independently from historical PV data [96] and primarily rely on the stability of prevailing weather conditions [97]. Hence, employing post-processing techniques proves to be an effective tool in this context [98]. NWP forecasting models work by solving mathematical formulations grounded in meteorological data. Consequently, their operational implementation requires substantial computational resources accompanied by elevated costs. Although NWP forecasting models have high computational demand, they exhibit enhanced reliability, particularly in the context of long-term weather forecasts [99].

As mentioned, NWP models are just one piece of the puzzle in physical methods for PV forecasting. PV module performance models account for the electrical behavior of the PV modules. They may consider factors like temperature, irradiance levels, and module characteristics to predict the power output, including inverter losses [100].

3.3.2. Statistical Methods

Statistical methodologies rely on establishing correlations between the explanatory variables employed and the model's target outcomes to facilitate precise forecasting. Time series models are the foundation for predicting the electricity generated by solar panels on buildings. A time series constitutes a sequence of observations of a parameter recorded at consecutive time intervals. The statistical model can model the system's behavior without needing to understand its underlying mechanisms [40].

With the advancement of science, Artificial Intelligence, which encompasses statistical models, is revolutionizing time series forecasting, offering powerful tools for forecasting PV generation, particularly because of the complex, nonlinear relationship between power output and weather conditions [101].

Artificial Neural Networks

ANNs are inspired by the structure and function of the human brain. The structure of an ANN typically includes an input layer, several hidden layers, and an output layer. It also incorporates connection weights, biases, an activation function, and a summation node. A schematic representation of Shallow Neural Networks [102] and Deep Neural Networks (DNNs) is provided in Figure 3 [103].

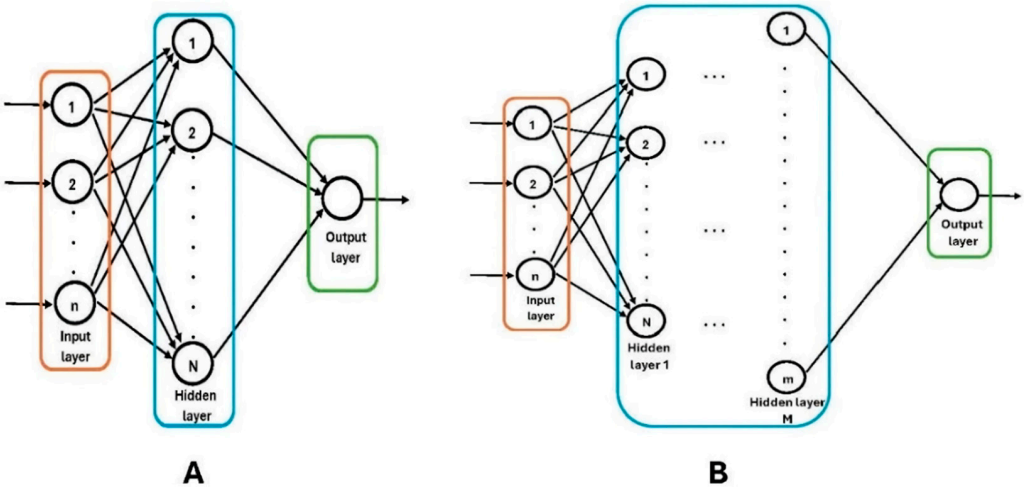


Figure 3. Shallow Neural Networks (**A**) and Deep Neural Networks (**B**).

Fundamentally, an ANN is an information processing model that is intrinsically structured to autonomously identify and monitor the relationships among various data. ANNs stand out as a suitable choice for forecasting PV generation due to their self-training capability, which involves continuously refining their predictions by comparing forecasted values to actual outcomes. Furthermore, ANNs' inherent self-learning ability to minimize future discrepancies between forecasted values and actual outcomes proves to be a crucial factor for accurate forecasting. This adaptability makes ANNs particularly well suited for forecasting PV generation [101]. While ANNs offer remarkable capabilities, they also have drawbacks, namely inaccurate predictions, high processing time, and no proven method for optimizing network structures [104].

Other Machine Learning Methods

The building sector applies a range of forecasting techniques, with ML being the most widely adopted approach. ML algorithms are inherently data-driven, meaning they rely on large datasets to extract patterns and insights [105]. Since PV generation often exhibits nonlinear dependencies on weather conditions [106], ML algorithms can efficiently process and make sense of complex and multi-dimensional data [107] generated by buildings, such as sensor data, meter data, and weather data. This allows ML models to capture the complex relationships between different factors that affect PV generation [108].

In addition, ML models can adapt to changing patterns. If there is a change in weather patterns, ML models can learn from the new data and adjust their forecasts accordingly [109]. This is important, because PV generation is constantly changing throughout the day and over time. ML models can analyze historical time series data, which consists of data points recorded over time at regular intervals [110]. This data format is particularly valuable, because it captures the temporal dependencies (the relationships among data points at different time intervals) and trends in PV generation [111]. By examining these patterns, ML models can provide accurate forecasts of future energy generation [112].

This capability allows building owners and operators to make more informed decisions regarding energy storage and demand side management strategies.

The process of developing a data-driven model for PV generation prediction involves four essential steps: data collection, data preprocessing, model training, and model testing [103]. Data collection entails acquiring historical PV generation and relevant data, such as outdoor weather conditions, which will be used in model training. Data preprocessing involves techniques such as data cleaning and data reduction, all of which aim to enhance the data quality and usability [113]. Figure 4 presents a step-by-step process for applying ML to solve real-world problems.

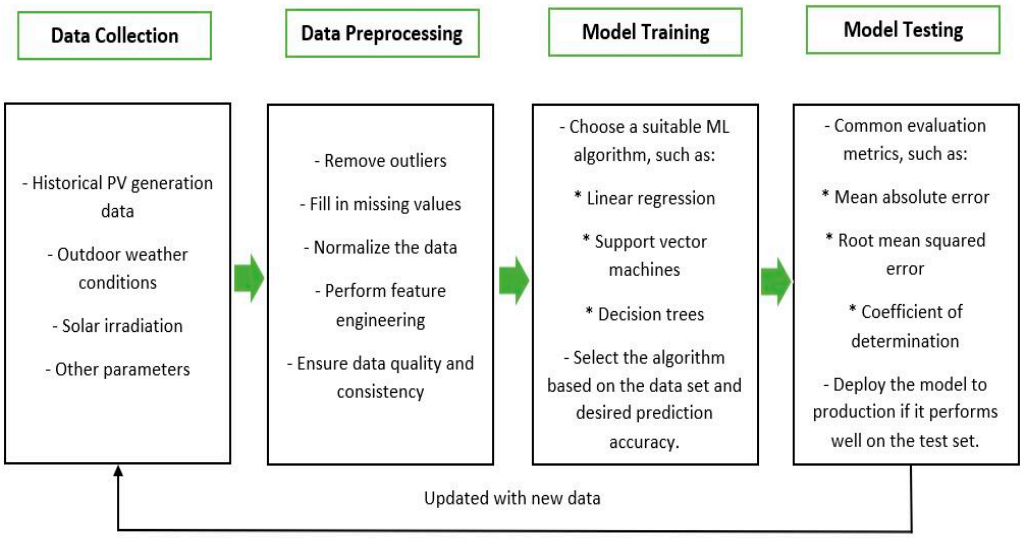


Figure 4. The step-by-step process for applying ML.

Sundararajan and Sarwat [114] conducted statistical analyses to understand missing data in PV systems within a grid. They assessed the efficacy of diverse imputation methods, such as K-Nearest Neighbors (KNNs), random imputation, multiple imputation, and Random Forests (RFs). The results suggested that KNNs and RFs emerged as the most effective techniques for imputation. Model training involves feeding the model with a training dataset, which enables the model to learn from the data and identify patterns [115]. Finally, model testing aims to assess the model's performance using standard evaluation measures, which ensures the model's reliability and accuracy [116].

Support Vector Machine (SVM) is one of the most important types of ML algorithms that draw upon statistical learning theory and the concept of structural risk minimization. SVMs have demonstrated superior accuracy and computational time compared to ANNs in tackling nonlinear problems [104].

Another important approach in ML to forecast PV energy generation is ensemble learning. Ensemble learning methods create multiple models and then combine them to produce improved results [117]. These methods are particularly effective in reducing overfitting, improving accuracy, and enhancing the stability and robustness of the model [118]. Ensemble methods can effectively handle the inherent variability of solar irradiance affecting PV generation [119]. Ensemble methods come in various forms, such as bagging, which trains models on different subsets of data (e.g., RFs), and boosting, which leverages the weaknesses of previous models to improve subsequent ones [120]. For instance, RFs, an example of a bagging technique, can be used to predict PV energy generation by creating a multitude of Decision Trees (DTs) at training time and outputting the class that is the mode of the classes of the individual trees [121].

In summary, the integration of ML in PV generation forecasting in buildings has demonstrated its potential to address the complexities of nonlinear relationships and temporal dependencies in energy data. By leveraging large and diverse datasets, ML models enhance forecast accuracy and adapt effectively to dynamic weather patterns. Additionally, advancements in methods such as ensemble learning further improve model performance. These strengths position ML as a powerful tool for PV generation forecasting, empowering stakeholders to make informed decisions that align with economic, technical, and environmental objectives.

3.3.3. Hybrid Approach

Hybrid methods combine two or more different types of models to leverage the strengths of each. In the case of accurate forecasting of renewable energy production and demand, the complexities associated with solar radiation and weather conditions pose a significant challenge for modeling using simple methods, particularly when high precision is crucial. Consequently, relying on a solitary model can result in significant errors. In recent years, hybrid models have been introduced to address the limitations of individual models and enhance forecasting precision [122].

3.3.4. Ensemble Approach

Ensemble methods combine multiple models of the same type to create a stronger overall model [123]. Although hybrid methods and ensemble methods share similarities, they have key differences. Their concept and differences are discussed in Table 3.

Table 3. The concept and differences between ensemble and hybrid methods.

Aspect	Ensemble Methods	Hybrid Methods
Definition	Combine multiple instances of the same or similar models [123] (e.g., RFs (ensemble of DTs))	Combine different types of models (e.g., Physical model + ML) [124]
Focus	Combine similar model outputs to create a stronger predictive model [123]	Captures different aspects of the problem (e.g., physical + stochastic patterns) [124]
Use in PV Forecasting	Effective for improving overall prediction accuracy	Modeling complex environmental and system interactions

3.4. Evaluation Approaches in PV Generation Forecasting for the Building Sector

This section explores evaluation approaches for PV generation forecasting, which are critical for maximizing the effectiveness and reliability of predictions.

3.4.1. Evaluation Metrics for PV Generation Forecasting in the Building Sector

Evaluation is an essential step for forecasting models, since it ensures the reliability of the forecasts and ultimately leads to more informed decision-making [125]. Some standard evaluation metrics that can be used for PV generation forecasting in buildings include:

- Mean Absolute Error (MAE) measures the average absolute difference between the predicted and actual values. It is primarily employed for model fitting, helping to select the optimal parameters for a given model. It is also utilized for model validation, model selection, comparing different models, and evaluating forecasts [126].
- Root Mean Squared Error (RMSE) measures the square root of the average squared difference between the predicted and actual values. RMSE is more sensitive to larger errors than MAE due to the squaring effect. When dealing with normally distributed errors, RMSE is a more suitable choice for evaluating model performance compared to MAE [126].

- Mean Absolute Percentage Error (MAPE) measures the average absolute percentage difference between the predicted and actual values. MAPE is a good metric due to its scale independency and interpretability. However, it can be problematic with zero actual values [127].
- Normalized Root Mean Squared Error (NRMSE) normalizes the RMSE by the range of the actual values, making it a more useful metric for comparing models with different scales of prediction [128].
- Coefficient of Determination (R^2) measures the proportion of variance in the actual values that is explained by the model. R^2 is a good measure of the model's fit to the data [129], but it has limitations with nonlinear models [130]. It is often used to evaluate the effectiveness of energy efficiency measures, model stability, and reliability [131].
- Mean Bias Error (MBE) measures the average difference between the predicted and actual values. This metric helps to identify systematic errors in the model, indicating whether the model tends to overpredict or underpredict [132].

The choice of evaluation metrics should be based on the specific needs of the application. In many research studies, common metrics such as MAE, RMSE, and MAPE [133] are used to evaluate quality in regression tasks. R^2 is also widely employed to assess energy efficiency measures in various research studies [134,135]. No single evaluation metric is perfect, and it is generally recommended to apply a combination of metrics to assess the performance of a forecasting model.

Table 4 provides a comprehensive analysis of the key metrics utilized. The diverse array of evaluation metrics equips researchers and practitioners with a robust toolkit to assess and enhance the performance of forecasting models for PV generation, ensuring a nuanced understanding of model effectiveness.

Table 4. Key Metrics.

Metric	Abbreviation	Formula	Explanation
Maximum Error	MaxE	$\max_{i=1}^{n}(\text{prediction}_i - \text{actual}_i)$	The maximum error between the predicted and actual values
Minimum Error	MinE	$\min_{i=1}^{n}(\text{prediction}_i - \text{actual}_i)$	The minimum error between predicted and actual values
Absolute Error	AE_i	$\mid \text{prediction}_i - \text{actual}_i \mid$	The absolute difference between predicted and actual values
Maximum Absolute Error	MaxAE	$\max_{i=1}^{n} \mid \text{prediction}_i - \text{actual}_i \mid$	The maximum of absolute errors
Minimum Absolute Error	MinAE	$\min_{i=1}^{n} \mid \text{prediction}_i - \text{actual}_i \mid$	The minimum absolute error
Maximum Absolute Percentage Error	MaxAPE	$\max_{i=1}^{n} \mid \frac{\text{prediction}_i - \text{actual}_i}{\text{actual}_i} \mid$	The maximum percentage error relative to actual values
Relative Error	RE_i	$\frac{\text{prediction}_i - \text{actual}_i}{\text{actual}_i}$	The error relative to actual values
Mean Relative Error	MRE	$\frac{1}{n}\sum_{i=1}^{n}\frac{\text{prediction}_i - \text{actual}_i}{\text{actual}_i}$	The average of relative errors
Relative Absolute Error	RAE	$\frac{\sum_{i=1}^{n}\lvert\text{prediction}_i - \text{actual}_i\rvert}{\sum_{i=1}^{n}\lvert\text{actual}_i - \overline{\text{actual}}^*\rvert}$	The error relative to the absolute deviation from the mean of actuals
Mean Absolute Error	MAE	$\frac{1}{n}\sum_{i=1}^{n} \mid \text{prediction}_i - \text{actual}_i \mid$	The average of absolute errors
Normalized Mean Absolute Error	NMAE	$\frac{\text{MAE}}{\text{actual}_{max} - \text{actual}_{min}}$	Normalizes MAE by the range of actual values

Table 4. *Cont.*

Metric	Abbreviation	Formula	Explanation
Mean Absolute Percentage Error	MAPE	$\frac{1}{n}\sum_{i=1}^{n}\left\lvert\frac{\text{prediction}_i-\text{actual}_i}{\text{actual}_i}\right\rvert$	The average of absolute percentage errors
Mean Squared Error	MSE	$\frac{1}{n}\sum_{i=1}^{n}(\text{prediction}_i-\text{actual}_i)^2$	Average of squared errors
Root Mean Square Error	RMSE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\text{prediction}_i-\text{actual}_i)^2}$	The square root of MSE
Root Mean Squared Percentage Error	RMSPE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\text{prediction}_i-\text{actual}_i}{\text{actual}_i}\right)^2}$	The square root of the mean squared percentage error
Normalized Root Mean Square Error	NRMSE	$\frac{\text{RMSE}}{\text{actual}_{max}-\text{actual}_{min}}$	Normalizes RMSE by the range of actual values
Relative Root Mean Squared Error	RRMSE	$\frac{\text{RMSE}}{\overline{\text{actual}}}$	RMSE relative to the actual mean
Coefficient of Variation	CV	$\frac{\text{StandardDeviationofactual}}{\overline{\text{actual}}}$	Measures the variability of actual values relative to their mean
Coefficient of Variation of Root Mean Squared Error	CVRMSE	$\frac{\text{RMSE}}{\overline{\text{actual}}}$	Normalizes RMSE relative to the mean of actuals
Mean Absolute Scaled Error	MASE	$\frac{\text{MAE}}{\frac{1}{n-1}\sum_{i=2}^{n}\lvert\text{actual}_i-\text{actual}_{i-1}\rvert}$	Scales MAE using the mean absolute error from a naive prediction model
Coefficient of Determination	R^2	$1-\frac{\sum_{i=1}^{n}(\text{actual}_i-\text{prediction}_i)^2}{\sum_{i=1}^{n}(\text{actual}_i-\overline{\text{actual}})^2}$	The proportion of variance in actual values explained by predictions
Mean Bias Error	MBE	$\frac{1}{n}\sum_{i=1}^{n}(\text{prediction}_i-\text{actul}_i)$	The average bias between predictions and actuals
Normalized Mean Bias Error	NMBE	$\frac{\text{MBE}}{\text{actual}_{max}-\text{actual}_{min}}$ or $\frac{\text{MBE}}{\overline{\text{actual}}}$	Normalizes the mean bias error by either the range or mean of actuals
Mean Absolute Deviation	MAD	$\frac{1}{n}\sum_{i=1}^{n}\lvert\text{actual}_i-\overline{\text{actual}}\rvert$	The average deviation of actual values from their mean

* $\overline{\text{actual}}$ is the mean.

3.4.2. Post-Processing and Quantifying Uncertainty for PV Generation Forecasting in the Building Sector

Post-processing and quantifying forecast uncertainty are key players in improving the performance of forecasting models. Post-processing involves applying various techniques to the model's forecast to improve its accuracy and interpretability [136]. Post-processing helps correct errors or biases in initial PV generation forecasts [137]. Solar PV generation is highly dependent on cloud cover and solar irradiance. Thus, these errors can be caused by inaccuracies in the forecasting model, weather data, or other factors. Post-processing can incorporate cloud cover data and advanced irradiance modeling to improve forecasting accuracy. By assimilating real-time cloud cover information, forecasts can be adjusted to reflect changing weather conditions [138]. In addition, PV generation can vary significantly based on the specific location of the building. Post-processing allows for the integration of site-specific data, such as shadowing effects, panel orientation, and local weather conditions, to enhance the accuracy of forecasts for a particular building [96].

In a practical demonstration, Böök and V. Lindfors [96] proposed a post-processing method for improving the accuracy of PV output forecasts by utilizing actual PV output data to adjust a baseline model that was trained on NWP forecasts. This approach was validated by comparing the performance of the baseline and adjusted forecasts against

actual PV output, demonstrating its ability to enhance the accuracy of forecasts for specific sites. Alvarenga et al. [139] analyzed three approaches (an ANN and a linear statistical model, along with Kalman filter and Kernel Conditional Density Estimation) aimed at enhancing the resolution of solar irradiance forecasts from NWP. Applying a combination of an ANN followed by Kalman filter post-processing resulted in a notable reduction of 45% in MAE and 91% in MBE.

In the context of PV generation forecasting in buildings, quantifying forecast uncertainty involves estimating the range of possible outcomes and their likelihood. Fluctuations in weather patterns make it challenging to accurately predict the amount of electricity produced by PV systems. Cloud cover, atmospheric conditions, and seasonal changes can impact the amount of sunlight reaching solar panels. Quantifying uncertainty involves considering variations in weather forecasts and their potential effects on PV generation [30]. In addition, inherent limitations of the models applied to forecast PV generation [140] and variability in sensor readings and calibration errors contribute to the overall uncertainty in solar energy output predictions [141]. The Monte Carlo method has become the favored approach for dealing with forecast uncertainty in PV generation and building energy analysis, outpacing other methods in terms of both popularity and effectiveness [142].

Post-processing and quantifying forecast uncertainty play crucial roles in data-driven forecasting models. These methods address prediction uncertainties in PV generation, enhancing accuracy, reliability, and strategic insights, ultimately supporting better decision-making in the building sector.

3.5. An Analysis of Building Attached and Building Integrated Photovoltaic

In this section, the distinct approaches of BAPV and BIPV methods for seamlessly incorporating solar arrays into building structures are explored. The discussion examines the impacts of the installation and integration nuances of BIPV and BAPV. In the following, the simulation challenges in the mentioned buildings are briefly explained. Finally, a review of relevant articles published in recent years is subsequently completed.

3.5.1. Introduction to BAPV and BIPV

Two methods for incorporating PV arrays into buildings are BAPV and BIPV. The primary distinction between BAPV and BIPV lies in their integration into respective buildings. In the BAPV approach, modules are installed onto existing surfaces by stacking them after the completion of the building. The integration of BAPV is less seamless with the architecture, as it often involves adding solar panels as an additional layer to the building. On the other hand, it is often more straightforward to implement in existing buildings. The BIPV method involves replacing conventional building components, such as roofing, windows, or façades, with materials containing solar modules. The main advantage of BIPV is its seamless integration into the building's design, contributing to both energy generation and architectural aesthetics.

3.5.2. Challenges in Simulating BIPV and BAPV Systems

Simulating building-attached and specially integrated PV systems poses some challenges, such as heterogeneous irradiance patterns and uneven temperature distribution across the PV modules. Higher temperatures can negatively impact solar cell efficiency, leading to reduced power output [143]. BIPV and BAPV systems' thermal boundary conditions diverge from those of conventional PV power plants. Conventional PV power plants are typically ground-mounted in open fields with good airflow, allowing for natural heat dissipation. In addition, conventional PV plants often have space between panels for ventilation. On the other hand, building materials like roofs can absorb and retain heat, further increasing the temperature of PV panels in buildings. BIPV and BAPV systems

may have limited ventilation due to their close integration with the building, potentially compromising power prediction accuracy. BIPV and BAPV modules frequently encounter partial shading and masking conditions (dirt, dust, or other debris covering the solar panels), which affects the generated energy [144].

3.6. PV Generation Forecasting in the Building Sector

This subsection reviews the studies conducted in the fields of BIPV and BAPV.

3.6.1. Review of BIPV Studies and Modeling Approaches

This section reviews various methodologies for BIPV generation forecasting, including physical approaches, statistical methods, hybrid techniques, and ensemble models. The studies discussed explore different modeling strategies, data sources, and evaluation metrics to assess prediction accuracy under diverse conditions.

Li et al. [145] developed a multi-physics model predicting electrical, thermal, and structural performance in various conditions, considering shading and masking. The model included a simplified electrical representation for simulating PV system power output under the mentioned conditions. Based on another physical approach called the Sandia Array Performance Model (SAPM), Polo et al. [146] utilized data from satellite-derived solar irradiance and a digital model derived from light detection and ranging information to estimate PV generation. An observed RMSE of 8% was obtained for the monthly predicted power for the west-facing façades.

Hao et al. [147] investigated the practical use of ANNs to assess the energy production of façades incorporating these technologies. Under realistic conditions, Nguyen and Ishikawa [148] employed an ANN to forecast the annual output energy of BIPV, considering the two-terminal perovskite/silicon tandem cells. The outcomes indicated that the suggested model achieved a MAPE of 0.01884 for the training set. Paiva et al. [149] presented a methodology comprising a data processing strategy and ANN modeling specifically designed for intra-day PV generation prediction. ANN exhibited superior performance compared to the persistence model used as the benchmark. By applying ANN, Polo et al. [150] formulated a model for PV generation forecasting. The methodology introduced involved integrating shading computation with ML algorithms tailored for PV cells. The MRE varied within the range of 6–15%.

In the domain of forecasting PV generation in buildings, estimating the Maximum Power Point (MPP) plays a crucial role in fault detection and diagnosis. Utilizing an ANN, Lee et al. [151] developed a model aimed at estimating the MPP for BAPVs. The validation results of the model suggested satisfactory performance, as evidenced by a CvRMSE measuring less than 30%.

Dimd et al. [152] employed Long Short-Term Memory (LSTM) to assess the influence of diverse orientations, including different tilts and azimuth angles, on the precision of a PV prediction model. In this scenario, the incorporation of mixed orientations led to a 51% increase in RMSE. With a similar goal, Dimd et al. [153] achieved a 34% reduction in RMSE by developing a forecasting model that considers all mixed orientations, i.e., different azimuth angles and tilt, since the factors substantially impact the prediction accuracy.

Jeong et al. [154] employed the Nonlinear Autoregressive with exogenous input (NARX) network as the forecasting model and modified Sky Condition (SC*). The model exhibited good performance, especially on overcast days, when utilizing the SC* and the solar altitude.

When colored PV modules are in use, predicting PV generation becomes more challenging due to the variability in shading loss, which is influenced by the entrance altitude of the irradiance. Shin et al. [155] developed a Regression Neural Network model to tackle

the aforementioned challenge. According to the outcomes, the suggested model exhibited superior performance compared to the Linear Regression (LR) model. Utilizing organic solar technology, Corrêa et al. [156] projected the output of PV systems through several LR models. The most successful models attained R^2 values of 0.76 and 0.81, corresponding to simple and multiple regressions, respectively.

Multiple algorithms were utilized in various papers to generate predictions, and the outcomes of their simulations, along with other relevant metrics, were compared. Utilizing statistical methods and ML techniques, Fara et al. [157] created both an Autoregressive Integrated Moving Average (ARIMA) model and an ANN for PV generation forecasting. A comparison of the mentioned model's outcomes revealed that the ARIMA model exhibited greater efficiency compared to the ANN model. Luo et al. [158] introduced three multi-objective prediction frameworks utilizing an ANN, SVM, and LSTM. These frameworks aim to simultaneously predict the electrical power output from BIPV, cooling, and lighting loads. The predictive model utilizing SVM required the shortest computation time, while the ANN-based model demonstrated the most favorable outcome with the smallest MAPE.

Kabilan et al. [159] estimated PV power generation through the application of ANN, Quadratic Support Vector Machine (QSVM), and DTs. Utilizing the LR coefficients enhanced the accuracy of the forecast. According to the results, the NN achieved the highest accuracy, with a RMSE of 4.42%. Dourhmi et al. [160] evaluated five methods to identify the most effective prediction approach: ANN, RF, Decision Tree Regression (DTR), Support Vector Regression (SVR), and LR. Evaluation metrics such as MAE, MSE, RMSE, and R^2 were employed to assess the accuracy of these methods. The results indicated that the proposed ANN and RF models yielded the most favorable outcomes.

Abouelaziz et al. [161] introduced a methodology that integrates photogrammetry with Deep Learning (DL) techniques to generate data-driven PV output forecasts. Their findings revealed that the Convolutional Neural Network (CNN) model achieved superior performance in prediction accuracy compared to other ML models. Lee et al. [162] applied ML methods for predicting PV generation through the incorporation of feature engineering. Among Recurrent Neural Networks (RNNs), ANNs, SVM, RF, Classification And Regression Trees (CARTs), and CHi-square Automatic Interaction Detection (CHAID), RNNs exhibited superior performance in predicting power at both hourly and daily resolutions. Feature selection, utilizing variable importance determined with SVM, was implemented by applying weather forecast data.

Jouane et al. [163,164] formulated three models employing LSTM, CNN, and a hybrid model (CNN-LSTM) within a Positive Energy Winter House context (houses generating more energy than their consumption). The evaluation of these models was conducted using MAE, RMSE, and MSE metrics. The distinction between the two articles is that [163] included side parameters such as temperature, while [164] relied solely on historical data on PV generation. While the CNN-LSTM model demonstrated better accuracy, it showed a slower performance compared to the others. Sarkar et al. [165] created a hybrid model for forecasting PV generation by combining RF, LSTM, and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN). The findings demonstrated that the hybrid strategy demonstrated superior performance compared to the reference methods.

Polo et al. [166] investigated two DT-based ML algorithms, namely RF and Extreme Gradient Boost (XGBoost), in their study. They observed that, in deterministic forecasting, RF generally exhibited a slightly superior RMSE compared to XGBoost across various cases.

As a summary of the section, Table 5 presents the methodologies employed in the reviewed articles on BIPV.

Table 5. Methodologies employed in the reviewed articles on BIPV.

Models		Methodologies/Techniques	Refs.
Physical Models	-	Simplified electrical mathematical model considering multi-physics	[145]
		SAPM	[146]
Statistical Models	NN	ANN	[147–151,157–162]
		LSTM	[152,153,158,163,164]
		NARX	[154]
		Regression Neural Network	[155]
		CNN	[161,163,164]
		RNN	[162]
	ML	LR	[156,160,161]
		ARIMA	[157]
		SVM	[158,162]
		QSVM	[159]
		DT	[159,161]
		DTR	[160]
		SVR	[160,161]
		CART	[162]
		CHAID	[162]
Hybrid Models	-	CNN-LSTM	[163,164]
		RF-LSTM-CEEMDAN	[165]
Ensemble Models	-	RF	[160–162,166]
		Adaptive Boosting	[161]
		XGBoost	[166]

An analysis of Section 3.5.1 shows that only two studies [164,165] on BIPV utilized only historical PV data as input variables. In contrast, the majority of the other studies relied primarily on weather data.

From Table 5, it can be concluded that statistical methodologies dominate because they are highly adaptable and effective at capturing the nonlinear relationships inherent in PV generation data. Techniques like Neural Networks and traditional methods of ML are frequently used, because they can handle time series data and adjust to dynamic weather conditions. In contrast, physical models, while based on system physics, require detailed input parameters that may be difficult to obtain in real-world scenarios. Hybrid and ensemble models, although powerful, are more complex and computationally demanding, which may limit their widespread application. Overall, the large presence of statistical models may reflect their historical utility, simplicity, and ease of implementation, but the increasing inclusion of Machine Learning, hybrid, and ensemble methods points to an evolution in the field toward more accurate, albeit complex, forecasting methodologies.

3.6.2. Review of BAPV Studies and Modeling Approaches

This section reviews several forecasting methodologies for BAPV power generation, including physical models, statistical approaches, hybrid and ensemble models, and emerging decentralized methods such as federated learning. The reviewed studies examine various modeling approaches and evaluation metrics to evaluate forecasting accuracy across different conditions.

Zhi et al. [167] introduced a physical model incorporating an enhanced Maximum Power Point Tracking algorithm, achieving a MAE of 15.9% under diverse weather conditions. Massucco et al. [168] devised a hybrid model, utilizing either clear sky models or an ensemble of ANNs, selecting the optimal approach via DTs.

Reynolds et al. [169] utilized a back-propagation ANN to forecast the next 24 h, achieving R^2 values of 0.9489 for training and 0.9412 for testing. Gheouany et al. [170] utilized a stacking-based ANN to forecast PV power output. They predicted weather variables, including global horizontal irradiance, temperature, and wind speed, as inputs for both long-term (30-day) and short-term (24-h) forecasting, significantly reducing the errors. Sabzehgar et al. [171] created a NN-based NWP model for a residential microgrid, which outperformed SVM and Multi-Variable Regression (MVR) in terms of MSE and MAPE accuracy.

Kaffash et al. [172] employed a Feed-Forward Neural Network (FFNN) for day-ahead PV generation forecasting, achieving a NRMSE of 11.89 without weather data. The same authors [173] introduced an ensemble method combining FFNN and SVM, using RF for feature selection. Despite the higher computational cost associated with SVR or ANN, the ensemble approach improved the forecasting accuracy.

Chung [174] introduced a multilayer FFNN to forecast next-day PV generation based on current weather conditions, showing that more input variables did not always improve accuracy. Costas et al. [175] utilized a Standard Neural Network (SNN) for PV generation and electricity demand forecasting, achieving a NRMSE of 7.15% (PV generation). Ghenai et al. [176] applied NARX to predict bifacial PV output, finding that increasing the surface albedo from 0.2 to 0.5 and 0.8 boosted the annual bifacial power generation by 7.75% and 14.96%, respectively.

Parvez et al. [177] developed a Multi-Layer Perceptron (MLP) model for rooftop PV forecasting, achieving MAE and NRMSE of 0.0809 and 0.0054, respectively. Sarmas et al. [178] suggested an online MLP model for PV output, improving MAE by 11.9% over offline models. Meteier et al. [179] utilized Extra-Trees (ETs) and MLP for energy production forecasting by incorporating weather- and time-based features. The MLP yielded the most precise prediction, achieving an R^2 score of 84.81%. Piotrowski et al. [180] examined ten forecasting methods (both hybrid and team approaches) and introduced a hybrid approach using three independent MLP-type neural networks, minimizing RMSE and nMAPE.

Rooftop PV systems are commonly located behind the meter and unmonitored. In addition, traditional centralized prediction methods raise privacy concerns (due to the need to share data with a centralized entity), prompting the need for decentralized methods like federated learning (FL). Hosseini et al. [181] developed a MLP-based FL model for behind-the-meter PV forecasting, achieving an 18.7% reduction in RMSE. Kim et al. [182] forecasted behind-the-meter production by considering the unauthorized PV capacity, employing the SVR and upscaling method. In comparison to estimates that do not incorporate the unauthorized PV capacity, the methodology results in a reduction of 2.95% and 5.41% in NMAE and NRMSE, respectively. González et al. [183] compared ANN and SVM using sky camera images and found similar error rates. VanDeventer et al. [184] introduced a stochastic Genetic Algorithm-based Support Vector Machine model for short-term prediction, boasting accuracy, speed, and memory efficiency over standard SVM.

Using historical PV generation and weather data, Bird et al. [185] employed SVM, RF, and ANN for prediction objectives, with RF achieving the best performance (MRE of 2.7%). Scott et al. [186] compared ML algorithms for university campuses, finding that RF had the lowest RMSE compared to SVM, LR, and NN. Teixeira et al. [187] estimated PV generation by applying ANN, SVM, RF, and ARIMA, with the appropriate hyperparameters tailored for each method. The best outcomes were achieved using ANN and SVM, respectively.

Tavares et al. [188] compared two Neural Network architectures, an ANN and a DNN, for PV generation in buildings, with the ANN model exhibiting superior forecasting accuracy. Duhirwe et al. [189] developed a four-layer DNN, employing the PV generation dataset. The R^2 values of the developed DL-based forecasting models exceeded 0.95.

Shivam et al. [190] developed a dual forecasting model using the Residual Dilated Causal Convolutional Network to forecast both energy generation and electric load, outperforming both SVM and ANN models in terms of accuracy. Kazem et al. [191] introduced four predictive models employing DL techniques, specifically utilizing Time Lagged Recurrent Networks and RNNs by considering input data such as solar irradiance and temperature.

Park et al. [192] employed single-LSTM and multi-LSTM models for their forecasting purposes, noting no significant difference, though the multi-layer model exhibited a slightly lower error rate. Son and Jung [193] formulated 22 multivariate models by integrating six meteorological factors. To forecast power, the LSTM method was employed, and the results showed that the PV power forecast was influenced by six meteorological factors, ranked from most to least important, including solar radiation, sunlight period, wind speed, temperature, cloud cover, and humidity. Xu et al. [194] introduced a LSTM-based model for rooftop PV output forecasting, incorporating the predictions into a Q-learning framework to optimize pricing for PV and ESS utilization.

Alden et al. [195] utilized an Encoder–Decoder LSTM for day-ahead PV output prediction, achieving goodness of fit of 0.975 on an hourly basis and a daily CVRMSE of 11.1%. To enhance the balance between supply and demand, Wen et al. [196] devised a Deep Recurrent Neural Network with LSTM to forecast aggregated energy demand and PV power output within a building community. Using real-world data, the model outperformed both MLP and SVM. Lateko et al. [197] introduced a stacking ensemble model, incorporating DNN, ANN, LSTM, SVR, and CNN, with an RNN as the meta-learner. The MRE demonstrated improvements in the proposed ensemble learner model over individual models.

Iao et al. [198] applied CNN-LSTM, BiLSTM, and Gated Recurrent Unit (GRU) for PV generation and load forecasting, employing KNN interpolation to fill missing values in historical PV power data. Among the models, BiLSTM with KNN interpolation yielded the lowest RMSE but required the longest training time. Expanding on this work, the same authors [199] integrated KNN and Generative Adversarial Networks for data interpolation and augmentation, formulating LSTM and Transformer-based forecasting models. The model achieved an RMSE of 4.603 kW for PV power forecasting.

Hu et al. [200] proposed a LSTM model enhanced with self-attention mechanisms, achieving high accuracy and adaptability for both short-term and long-term forecasts. Hernandez-Robles et al. [201] investigated five Neural Network architectures, including the MLP, single-layer, dual-layer, and bidirectional LSTM models, as well as GRU. While the MLP model had the fastest computation times, the dual-layer LSTM model demonstrated the highest reliability and accuracy. Cordeiro-Costas et al. [202] illustrated the effectiveness of a hybrid LSTM-MLP model, emphasizing the advantages of NSGA-II for hyperparameter optimization and energy management in sustainable building practices, with a NRMSE of approximately 5.5%.

Li et al. [203] addressed the issue of incomplete historical and real-time measurement data by proposing a Recursive LSTM model for PV generation forecasting in buildings. The method outperformed the Gaussian Processes and ANN methods, indicating its capability to learn from time series PV data. Costa [204] evaluated LSTM, CNN, and Convolutional-LSTM (ConvLSTM) networks at half-hourly, daily, and monthly intervals, finding that CNN and ConvLSTM consistently achieved better performance than the Prophet algorithm and LSTM based on MAE, RMSE, and NRMSE metrics.

Sarmas et al. [205] applied meta-learning to four LSTM variants (Stacked LSTM, Bidirectional LSTM (BiLSTM), CNN-LSTM, and ConvLSTM), enhancing performance by up to 5% over the best base model. Guo et al. [206] introduced a hybrid model integrating Singular Spectrum Analysis and BiLSTM, optimized via the Bayesian Optimization algorithm.

The 7.5 and 15 min-ahead predictions by the model reduced errors by up to 380.51% and 296.01%, respectively.

Pan et al. [207] introduced a PV generation prediction approach and decoupling PV load in a residential power distribution system with behind-the-meter PV utilizing the KNN algorithm. The RAE values exhibited reductions of 31.60%, 44.08%, 42.40%, and 60.51% during spring, summer, autumn, and winter. Sangrody et al. [208] developed three Similarity-Based Forecasting Models (SBFMs)—basic SBFM, categorical SBFM, and hierarchical SBFM (hSBFM)—leveraging the KNN approach for PV generation forecasting at a 5-min resolution. Evaluated against ANN and persistence models, the results demonstrated that hSBFM exhibited the best overall performance.

Tripathy et al. [209] developed a Multi-Time Instant Probabilistic PV Generation forecasting model for rooftop PV installations, utilizing Quantile Regression Forests (QRFs), which outperformed basic Quantile Regression (QR). They later introduced a Quantile K-Nearest Neighbors Regression Averaging model [210] that achieved lower forecast errors compared to methods such as regression bootstrapping, traditional QR averaging, QRF, Quantile KNN, and basic QR.

To estimate behind-the-meter PV generation, Kabir et al. [211] combined a statistical demand prediction model with a physical PV output forecasting model using Hidden Markov model regression, reducing MSE by 44% compared to the consumer mixture model. They further utilized a Mixed Hidden Markov Model [212] to estimate the technical parameters of solar PV systems.

Allouhi [213] proposed a multi-objective optimization approach using a Genetic Algorithm (GA) to balance economic and environmental factors, incorporating Multiple Linear Regression (MLR) to forecast PV generation, achieving a NRMSE of 0.943. Kallio and Siroux [214] employed MLR and ANN methods, improving the accuracy of predictions by incorporating micro-inverter technology into their modeling approach. Capotosto et al. [215] compared persistence, MLR, and ARIMA for day-ahead PV generation forecasting in residential and public buildings, finding that ARIMA and MLR performed similarly for residential users, while MLR outperformed ARIMA in public buildings, highlighting that those sophisticated methods may not always yield the optimal solution.

Jiranantacharoen and Benjapolakul [216] employed ARIMA with the Kalman filter algorithm for five-minute intervals and day-ahead forecasting. Gellert et al. [217] assessed the ARIMA and Trigonometric Seasonal, Box-Cox Transformation, ARMA residuals, Trend, and Seasonality (TBATS) methods to forecast PV energy output in smart homes, with TBATS significantly outperforming the Neural Network-based methods.

Lee et al. [218] introduced Automatic Machine Learning (AML) to generate multiple forecasting models using meteorological data and solar power generation, employing 24 different algorithms. Furusawa et al. [219] utilized PyCaret, an AML tool, incorporating diverse ML methods like ET regressor, Ridge regression, Bayesian Ridge, and RF Regressor. The ET regressor yielded the best performance among the methods.

Lee et al. [220] devised a PV generation forecasting framework based on the autoregressive model, incorporating temperature, relative humidity, cloud cover, and solar radiation obtained from the ASHRAE Clear-Sky model, achieving a MAPE of 0.227 for sunny days and 0.386 on cloudy days. Bottieau et al. [221] developed cross-learning forecasting for household PVs, combining Gradient-Boosted Regression Trees, several NNs, and the KNN algorithm. Results demonstrated the superior performance of cross-learning forecasting over a clear sky-based physical approach.

Zhang et al. [222] developed a data-driven PV generation forecasting model using Spatial-Temporal Correlation Analysis within a Bayesian Network framework. The proposed method outperformed various models, including ARIMA, KNN, LSTM, Persistent

Model, the deterministic Autoregressive-based spatiotemporal forecasting algorithm, and an Autoregressive with exogenous input-based spatiotemporal solar forecast model. Bot et al. [223] implemented an ensemble of Radial Basis Function ANNs (RBF-ANNs) to predict variables relevant to HEMSs, particularly for PV power output forecasting. This study utilized a Multi-Objective Genetic Algorithm for both feature and topology selection, along with a stochastic convex-hull algorithm known as ApproxHull for data selection. Yang et al. [224] introduced a Competitive Swarm-Optimized (CSO) RBF-ANN, demonstrating superior accuracy in comparison to other benchmarks.

Sun et al. [225] introduced a WTEEMD-FCM-IGWO-LSTM approach for PV generation forecasting in buildings. The method integrated Wavelet Threshold Improvement Ensemble Empirical Mode Decomposition (WTEEMD) to reduce noise in weather data, followed by an adaptive weight-improved Fuzzy C-Means (FCM) clustering algorithm, was employed to categorize varying weather conditions. The Improved Gray Wolf Optimization (IGWO) algorithm was then employed for hyperparameter tuning, addressing suboptimal issues. Comparative analysis confirmed the approach's superior performance in PV power forecasting. Peng et al. [226] introduced an approach that utilizes Variational Mode Decomposition (VMD), an improved version of Informer, and an Enhanced Chaos Game Optimization algorithm. Significantly, the Informer model has been enhanced through the incorporation of Locality-Sensitive Hashing attention. Results indicated that VMD effectively reduced noise in PV power data, improving model accuracy. Khan et al. [227] introduced an improved generally applicable stacked ensemble algorithm (DSE-XGB) that incorporates ANN and LSTM for predicting PV generation. Performance comparisons against individual bagging, ANN, and LSTM demonstrated that DSE-XGB improved the R^2 value by 10–12% over competing approaches.

Etxegarai et al. [228] compared four distinct models. The first model was analytical, while the other three were ML-based, specifically SVR, NARX, and FFNN. SVR emerged as the model with the most precise forecasts. Kaffash et al. [229] investigated various forecasting models for scheduling a multi-energy system, including Persistence, Simple Moving Average, Least Absolute Shrinkage and Selection Operator (LASSO), FFNN, and SVR, to identify the most accurate PV generation forecasting methodology. LASSO exhibited the best performance across all metrics, whereas SVR performed the least satisfactorily. The study highlighted that, despite the complexity of models like FFNN and SVR, they were unable to surpass LASSO, primarily due to the limited dataset size, which constrained the training of intricate methods like FFNN. Hatamian et al. [230] explored PV power output prediction by using weather data and various ML models, including RF, XGBoost, KNN, MLP, and SVR. The analysis revealed similar forecasting patterns among MLP, KNN, and SVR, while RF and XGBoost exhibited comparable performance. Giovanni et al. [231] evaluated four ML algorithms—KNN, DTs, Support Vector Regression, and RFs—to predict daily power production. Among these, RFs emerged as the most effective model.

Costas et al. [232] evaluated ML (XGBoost, RF, and SVR) and DL (RNN, SNN, and CNN), identifying SVR, SNN, and CNN as the most fitting models for forecasting PV generation. Mohana et al. [233] employed several ML methods (RF, LASSO, SVM, LR, XGBoost, Polynomial Regression, and DL) and employed a backward feature elimination technique to identify the most relevant set of features. The mentioned method aided in revealing that similar outcomes can be achieved with a reduced set of features. Their results showed that the DL technique yielded the lowest error using the minimal set of selected features, while LR had the highest errors.

As a summary of the section, Table 6 presents the methodologies employed in the reviewed articles on BAPV.

Table 6. Methodologies employed in the reviewed articles on BAPV.

Models		Methodologies/Techniques	Refs.
Physical Models	-	An equivalent circuit model	[167]
		Clear-Sky Model	[168]
		Corrected Clear-Sky Model	[168]
		OpenModelica	[228]
Statistical Models	NN	ANN	[169–171,183,185–188,214]
		FFNN	[172,174,228,229]
		SNN	[175,232]
		NARX	[176,228]
		MLP	[177–180,201,230]
		DNN	[188,189,233]
		Residual Dilated Causal Convolutional Network	[190]
		RNN	[191,232]
		Time Lagged Recurrent Networks	[191]
		LSTM	[192–195,199,201,204]
		BiLSTM	[198,201]
		Deep Recurrent Neural Network-LSTM	[196]
		GRU	[198,201]
		Transformer models	[199]
		Recursive LSTM	[203]
		CNN	[204,232]
	ML	Persistence model	[180,215,229]
		LR	[180,186,233]
		KNN	[180,207,230,231]
		SVM	[180,183,185–187,233]
		SVR	[182,228–232]
		ARIMA	[187,215,217]
		SBFMs based on KNN	[208]
		QRF	[209]
		Quantile K-Nearest Neighbors Regression Averaging	[210]
		Markov model regression	[211,212]
		MLR	[213–215]
		TBATS	[217]
		AML	[218,219]
		Autoregressive model	[220]
		cross-learning	[221]
		Bayesian Network With Spatial-Temporal Correlation Analysis	[222]
		Simple Moving Average	[229]
		DTs	[231]
		LASSO	[229,233]
		Polynomial Regression	[233]
Hybrid Models	-	Connection of three MLP models	[180]
		MLP-based FL	[181]
		GA-SVM	[184]
		LSTM with self-attention mechanisms	[200]
		LSTM-MLP	[202]
		CNN-LSTM	[198,204]
		Singular Spectrum Analysis + BiLSTM + Bayesian Optimization	[206]
		ARIMA combined with the Kalman filter	[216]
		CSO-RBF	[224]
		WTEEMD-FCM-IGWO-LSTM	[225]
		VMD-Enhanced Chaos Game Optimization-Locality Sensitive Hashing Attention-Informer model	[226]

Table 6. *Cont.*

Models		Methodologies/Techniques	Ref.
Ensemble Models	-	Basic Ensemble Method of NNs	[168]
		Ensembles of FFNN and SVM	[173]
		ETs	[179]
		Gradient-Boosted Trees	[180]
		RFs	[180,185–187,230–233]
		Weighted Averaging Ensemble	[180]
		Ensemble Method with the RNN Meta-Learner	[197]
		Meta-learning used four models of LSTM	[205]
		RBF-ANN	[223]
		DSE-XGB	[227]
		XGBoost	[230,232,233]

By examining input variables for prediction models, it was observed that only 12 studies on BAPV incorporated historical PV data as input variables. By contrast, most other studies predominantly relied on weather data.

Statistical methods, especially Neural Networks such as ANN, MLP, and LSTM, are the dominant approaches in forecasting BAPV power generation. The literature favors these models for their strong capability to effectively capture and model complex, nonlinear patterns in the data. LSTM emerges as the most frequently DL-applied technique, attributed to its robustness in handling time-dependent data and generalization capability. Additionally, widely used ML methods like SVR and SVM have been employed as benchmark models. In contrast, physical models, while reliable in deterministic settings, are less prominent.

Hybrid models highlight the integration of strengths across methodologies, combining features like optimization (e.g., GA-SVM) and multi-model architectures (e.g., CNN-LSTM). These approaches address specific forecasting challenges, such as capturing long-term dependencies and improving prediction accuracy under variable conditions. Similarly, ensemble models have gained traction by leveraging diverse algorithms to reduce bias and variance, with RFs and XGBoost being especially notable.

Table 7 presents the metrics employed in the reviewed articles on BAPV, BIPV, and Total. Only metrics used at least five times are included in the table.

Table 7 highlights the prevalence of performance metrics in the forecasting of PV generation in buildings, with RMSE being the most commonly used metric. RMSE's popularity stems from its ability to heavily penalize larger errors, which is crucial in PV forecasting, where significant deviations can impact energy management. It is easy to interpret, as it is expressed in the same units as the predicted values, making it practical for comparing forecast accuracy. Other metrics, such as MAE, R^2, and MAPE, are also frequently used but serve slightly different purposes. MAE, for example, treats all errors equally and is often used to measure overall accuracy without prioritizing larger errors. R^2 helps understand the model fit, while MAPE normalizes errors by expressing them as percentages. However, their usage is less widespread compared to RMSE, likely because they do not penalize large errors as strongly.

This trend may stem from the fact that weather data can significantly enhance model accuracy by capturing external factors. However, the challenge lies in obtaining reliable real-time weather data, which are not always easily accessible or consistent across regions. Therefore, some studies may have chosen to rely solely on historical PV data. Among predicted climate variables, the Normative Sky Index shows the lowest correlation with PV power output under most sky conditions, making it redundant for forecasting [162]. Solar power forecasting is influenced by six key weather factors, ranked by importance: solar radiation, sunlight, wind speed, temperature, cloud cover, and humidity [186,193].

Table 7. Metrics employed in the reviewed articles on BAPV, BIPV, and Total.

Metric	BAPV	BIPV	Total
RMSE (37) + (17)	[168,170,174,177–181,184–187,191–193,196,198–201,204–206,209,210,214,216,219,222–228,230,231]	[145–150,152,153,155,157,159,160,162–166]	54
MAE (30) + (11)	[167,170,172,174,177,178,187,188,190,193,196,197,200,204,206–210,214,215,217–219,225–228,230,231]	[147–149,152,155,157,159,160,163–165]	41
R^2 (17) + (10)	[169,179,189–191,197,200,201,214,218,219,225–228,230,232]	[150,151,154–157,159–161,165]	27
MAPE (15) + (7)	[171,181,182,184–187,190,191,196,198,199,215,220,230]	[148,154,157–159,162,165]	22
NRMSE (21) + (0)	[168,170,172,175,177,178,182,183,188–190,197,202–204,208,213,216,221,229,232]	-	21
MSE (11) + (9)	[171,176,191,198,199,211–214,226,233]	[162,165,167,171,178,179,181–183]	20
NMAE (5) + (1)	[170,177,182,215,229]	[166]	6
MRE (5) + (1)	[167,173,185,197,208]	[150]	6
MBE (3) + (2)	[168,174,180]	[150,166]	5
NMBE (5) + (0)	[175,183,189,202,232]	-	5

Table 8 presents an overview of the reviewed studies on PV forecasting in the building sector. The table summarizes forecasting methodologies; evaluation metrics; deployment types (e.g., residential, office, and campus); and geographic distribution of the datasets. Additionally, the Tips column offers insights into partial shading, dust accumulation, building-specific PV patterns, or any other relevant information related to the article. While multiple prediction methods were used in some studies, the one with the lowest error is listed as a reference in the Method column.

Table 8. Overview of the reviewed PV forecasting studies in the building sector.

Methods	Forecasting Accuracy	Tips	Deployment Types	Geographical Location	Ref.
Simplified electrical mathematical model considering multi-physics	MSE & RMSE Illustrated on fig.	- Shading - Masking	University BIPV lab	Tianjin/China	[145]
SAPM	R^2 for each façade East = 0.73 South = 0.89 West = 0.90	Shading	University campus	Madrid/Spain	[146]
ANN	R^2 = from 0.63 to 0.88	Vertical Farming	Residential	Singapore	[147]
ANN	MSE = 1.26	_	Rooftop	Gifu/Japan	[148]
ANN	For 15-min MAE = 34.17 W/m^2 RMSE = 60.08 W/m^2	_	Engineering School	Goiania/Brazil	[149]
ANN	Based on the different façades MRE ranged approximately from 6% to 15%	Partial shading	University area	Madrid/Spain	[150]

Table 8. *Cont.*

Methods	Forecasting Accuracy	Tips	Deployment Types	Geographical Location	Ref.
ANN	R^2 = 0.9768 CvRMSE (%) = 35.22	Partial shading	_	Republic of Korea	[151]
LSTM	RMSE (kW) = 2.24 MAE (kW) = 1.12 WAPE (%) = 4.66	_	ZEB laboratory	Trondheim/Norway	[152]
LSTM	RMSE (kW) = 2.242 WAPE (%) = 4.664	_	ZEB laboratory	Trondheim/Norway	[153]
NARX	For Partially Cloudy days MAPE = 11.55% R^2 = 0.95	_	Office	Seoul/Korea	[154]
Regression Neural Network	RMSE (kW) = 0.0754 MAE (kW) = 0.0372	- Colored PV - Shading	_	Daejeon/Korea	[155]
Mathematical modeling with multiple regression	R^2 = 0.81	Organic PV	_	São Paulo/Brazil	[156]
ARIMA + ANN	Monthly: rRMSE (%) = from 6.2 and 53	_	University lab	Bucharest/Romania	[157]
ANN	MPE (%) = 6.29	Shading	Office building	The UK	[158]
ANN	RMSE = 4.42% R^2 = 0.8833	_	Residential buildings	Kovilpatti/India	[159]
CNN	MSE (kW) = 0.046 R^2 = 0.96	- Several models examined	Campus building	Strasbourg/France	[161]
RNN	MAPE (%) = 23.79	_	Office building	South Korea	[162]
CNN + LSTM	MSE & RMSE Illustrated on fig.	_	Winter house	Poschiavo/ Switzerland	[163]
CNN + LSTM	MAE = 4.98 RMSE = 14.06	_	Winter house	Poschiavo/ Switzerland	[164]
CEEMDAN + RF + LSTM	For a Flat roof: RMSE = 2.97 MAE = 2.475	_	Institute of Engineering	India	[165]
XGBoost	RMSE = 0.89 and 0.21 kW for south and east	Shading	Office	Madrid/Spain	[166]
Maximum Power Point Tracking algorithm	MAE =15.9% MRE = 18.1%	_	Household	Ruicheng/China	[167]
Hybrid Tree	RMSE (kW) = 1.5583 nRMSE = 0.3892	Shading	University/Economics School	Genova/Italy	[168]
Back-propagation ANN	R^2 (Train) = 0.9489 R^2 (Test) = 0.9412	_	Several types	Cardiff/The UK	[169]
Stacking-based ANN	Short-term: NRMSE (%) = 6.49 NMAE (%) = 3.68	_	Residential	Morocco	[170]
NN-based NWP	MAPE = 45.30%	_	Residential	San Diego/ The USA	[171]
FFNN	MRE = 9.15%	_	Research center	Genk/Belgium	[172]
FFNN + SVM + RF	NRMSE (%) = 11.89	_	Research center	Genk/Belgium	[173]
Multilayer FFNN	RMSE (kWh) = 1.421 MAE (kWh) = 1.133	_	_	Seoul/Korea	[174]

Table 8. *Cont.*

Methods	Forecasting Accuracy	Tips	Deployment Types	Geographical Location	Ref.
SNN	nRMSE = 7.15% nMBE = −0.21%	–	University Campus	Vigo/Spain	[175]
NARX	For Albedo $\alpha = 0.8$ nMSE = 6.06634×10^{-1}	Bifacial PV	_	Sharjah/UAE	[176]
MLP	MAE = 0.0809 nRMSE = 0.0054	–	Smart home (NREL Database)	USA	[177]
MLP	MAE = 6.697 RMSE = 13.260 nRMSE = 0.527	_	Headquarters	Terni/Italy	[178]
MLP	R^2 = 84.81 RMSE = 0.36	_	Smart Home	Swiss	[179]
MLP	RMSE (w) = 61.633 nMAPE (%) = 0.805	–	University PV Laboratory	Warsaw/Poland	[180]
MLP-based FL	Case 1 Average: RMSE (%) = 9.72 MAPE (%) = 12.88	Behind-the-meter	Several residences	New Mexico/USA	[181]
SVR	nMAE (%) = 2.95% nRMSE (%) = 5.41%	Behind-the-meter	Utility	Sydney/Australia	[182]
ANN	nRMSE (%) range from 7.71 to 21.43	- Several conditions of the sky	Solar Energy Research Centre	Almería/Spain	[183]
GA-based SVM	RMSE (W) = 11.226 MAPE (%) = 1.7052	–	Deakin University, Engineering Dep.	Victoria/Australia	[184]
RF	MRE (%) = 2.7	_	Several commercial rooftops	The UK	[185]
RF	RMSE = 32	_	A non-domestic building	The UK	[186]
ANN	MAPE from 0.1868 to 0.2073	–	GECAD research group building	Porto/Portugal	[187]
ANN	MAE (kW) = 0.09223 SMAPE = 0.04947 WAPE = 0.09894 nRMSE = 0.06213	_	A Residential Building (15 apartments)	Due to restrictions, it's not possible to tell the location	[188]
4-layer DNN	R^2 = 0.95	_	Retail shop	Korea	[189]
- Residual Dilated Causal Convolutional Network	R^2 = 0.9308 MAPE = 3.4819 SMAPE = 1.2003 NRMSE = 0.0589	–	Residential	Tainan/Taiwan	[190]
- Time Lagged Recurrent Networks and RNNs	MAPE (%) = 1.5032	_	Solar energy lab, Sohar University	Oman	[191]
Multi-layer-LSTM	Cv(RMSE) = 13.2 %	_	_	Jincheon/Korea	[192]
- LSTM - 22 multivariate models (combining solar radiation, sunlight, etc.)	Medium-term No. RMSE = 5.42 MAE = 3.21 Long-term No. RMSE = 9.23 MAE = 5.83	- 22 multivariate models (combining solar radiation, sunlight, etc.)	An industrial building	Gyeonggi-do/Korea	[193]

Table 8. *Cont.*

Methods	Forecasting Accuracy	Tips	Deployment Types	Geographical Location	Ref.
LSTM	_	_	- 6 apartments Each has 60 households	Australia	[194]
LSTM	Daily: CVRMSE (%) = 11.1	–	Smart home	Florida/USA	[195]
DRNN-LSTM	RMSE = 7.536 MAE = 4.369 MAPE (%) = 15.87	_	Residential	Yulara/Australia	[196]
Stacking ensemble models with RNN	MRE (%) = 4.29% nRMSE (%) = 6.16 MAE (kW) = 8.59 R^2 = 0.86	_	Industrial Co.	Taiwan	[197]
BiLSTM with KNN	RMSE (kW) = 1.984	_	NZEB Institute of Building Research	Shenzhen/China	[198]
GAN + KNN + LSTM + Transformer	Average reduction of RMSE (kW) = 4.603 in each LSTM-based	_	NZEB Institute of Building Research	Shenzhen/China	[199]
LSTM + self-attentions	RMSE (kW) = 0.651 MAE (kW) = 0.306 R^2 = 0.934	_	Houses	Urayasu/Japan	[200]
Dual-layer LSTM	RMSE = 0.0542	- Under cloudiness - Under solar intermittency	ZEB	Salamanca/Mexico	[201]
LSTM-MLP-NSGA II	NRMSE (%) = 5.5	_	University Campus	Vigo/Spain	[202]
Recursive LSTM	Singapore: nRMSE (%) =15.25 WMAPE (%) = 68.47 Australia: nRMSE (%) =15.12 WMAPE (%) = 38.95	- Under the missing data condition	- University Campus - University building	Queensland/ Australia Nanyang/ Singapore	[203]
Convolutional-LSTM	For Half an hour: MAE = 2.9 RMSE = 5.2 NRMSE = 0.03	–	Household	Sydney/Australia	[204]
Meta-learning + LSTM variants	Joao's location: RMSE = 2.273 Forecast skill index = 0.5	_	- Institution of social solidarity - Dental medicine unit - Elementary school	Lisbon/Portugal	[205]
Singular Spectrum Analysis + BiLSTM + Bayesian Optimization	15 Min-Ahead, Dataset 1: MAE = 9.44 RMSE = 12.29 R^2 = 0.971	–	Real-world rooftop stations	Eastern China	[206]
Utilizing the KNN algorithm	Spring season: RAE (%) = 30.61 MAE (kW) = 27.79	- Behind the Meter - Temperature Correction	Residential	Australia	[207]
hSBFM + KNN	MAE (W) = 826.2 MRE (%) = 15.3 nRMSE (%) = 10.8	_	University building	New York/The USA	[208]

Table 8. *Cont.*

Methods	Forecasting Accuracy	Tips	Deployment Types	Geographical Location	Ref.
QRF	First PV system: MAE = 0.666 RMSE = 0.924	_	Three nearby locations, rooftop PV	The USA	[209]
- Quantile KNN Regression Averaging	First PV system: MAE = 0.587 RMSE = 0.884	_	Three nearby locations, rooftop PV	The USA	[210]
- Hidden Markov Model Regression	MSE = 0.23 MASE = 3.08 CV = 0.62	Behind the Meter	Street	Austin/The USA	[211]
- Mixed Hidden Markov Model Regression	MSE = 0.13 MASE = 2.13 CV = 0.47	Behind the Meter	Street	Austin/The USA	[212]
MLR + GA	NRMSE = 0.943 MSE = 0.0049 NMSE = 0.9967	Soiling losses	- Commercial Buildings	Morocco	[213]
ANN	Overall: MAE = 13.34 MSE = 1517 RMSE = 38.96 R^2 = 0.935	Microinverter	- INSA ICUBE Laboratory	Strasbourg/France	[214]
MLR	Hour-Ahead: MAE (kW) = 1.42 MAPE (%) = 21.30 NMAE (%) = 4.37	–	- A residential - A public building	Rome/Italy	[215]
ARIMA + Kalman filter	RMSE = 0.135 nRMSE = 0.1688 Skill Score = 0.788	_	- Three PV rooftops located near each other	Thailand	[216]
TBATS	MAE (W) = 73.62	_	Smart homes	Romania	[217]
AML	Bayesian Ridge, PV1: MAE = 0.207 R^2 = 0.997	AML employs 5 algorithms	- ZEB - Energy Institute	South Korea	[218]
AML	ET regressor, Dec. MAE = 1.48 RMSE = 2.197 R^2 = 0.857	AML employs 18 algorithms	University Campus	Toyama/Japan	[219]
Autoregressive model	Sunny days: MAPE = 0.167 Cloudy days: MAPE = 0.327	- Under sky conditions	–	Korea	[220]
- Improved ensembled cross-learning forecasting	Sites Average (GBRT model): NRMSE (%) = 10.67	- Several models examined	Residential	- Western Belgium - Northern France	[221]
- Spatial-Temporal Correlation Analysis + Bayesian Network	15 min ahead: RMSE = 0.04	- 5 models examined	- More than 20 buildings	Australia	[222]
RBF-ANN + GA	Model 4: RMSE = 1.69	_	Residence	Algarve, Portugal	[223]
RBF-ANN + CSO	System 1: RMSE = 4.988×10^{-3}	–	- A community of building	Netherlands	[224]

Table 8. *Cont.*

Methods	Forecasting Accuracy	Tips	Deployment Types	Geographical Location	Ref.
WTEEMD + FCM + IGWO + LSTM	Average: MAE = 1.7915 RMSE = 2.1542 R^2 = 0.9915	- Several models examined	ZEB	Alice Springs/Australia	[225]
VMD + Informer + Enhanced Chaos Game Optimization	April: MSE (kW) = 224.21 MAE (kW) = 10.246 RMSE (kW) = 14.97 R^2 = 0.9732	- Several models examined	- Steel Structure Engineering Co.	Nantong/China	[226]
DSE-XGB	Case study (II): MAE (kWh) = 0.59 RMSE (kWh) = 0.78 R^2 = 0.96	–	Commercial Buildings	Netherlands	[227]
SVR	July: R^2 = 0.934 RMSE = 252.46 MAE = 141.81	- Four distinct models comparison	University building	Bidart/France	[228]
LASSO	NRMSE (%) = 14.24 NMAE (%) = 10.68 R^2 = 0.57	- 5 models examined	Industrial	Germany	[229]
RF	30 min: MAPE = 5.27 MSE = 0.032 RMSE = 0.110 R^2 = 0.987	- Several models examined	Smart buildings	Estonia	[230]
RF	MAPE (%) = 28.34 RMSE (kWh/d) = 139.10	- Shading - 4 models examined	Office and warehouse	Abruzzo/Italy	[231]
RF	nMBE (%) = 0.41 nRMSE (%) = 1.88 R^2 = 0.99	- Several models examined	- A single-family dwelling	Maryland/USA	[232]
PR	MSE is illustrated in the figure.	- Several models examined	Residential	Saudi Arabia	[233]

A column-by-column analysis and discussion of the above table is presented below.

Although RMSE and MAE (Tables 7 and 8) are widely used in the PV forecasting literature, these absolute error metrics are scale-dependent and therefore not directly comparable when studies involve systems with different installed capacities or generation patterns. The distribution of the data and the presence of outliers disproportionately affect RMSE (because errors are squared), and when the data range is large (e.g., from zero to very high irradiance values), RMSE can appear large even when relative errors are small. For these reasons, reporting RMSE/MAE alone can be misleading for cross-study or cross-system comparisons. Therefore, it is recommended to report capacity-normalized metrics such as NRMSE, NMAE, as well as CVRMSE, since these indicate the error per unit installed capacity or relative to mean production and substantially improve comparability across system sizes, climates, and aggregation levels. Finally, the chosen normalization method should be reported explicitly (for example, specifying that Prated is given in kWh).

Figure 5 displays the distribution of the reported NRMSE (%) values extracted from the surveyed studies (N = 15).

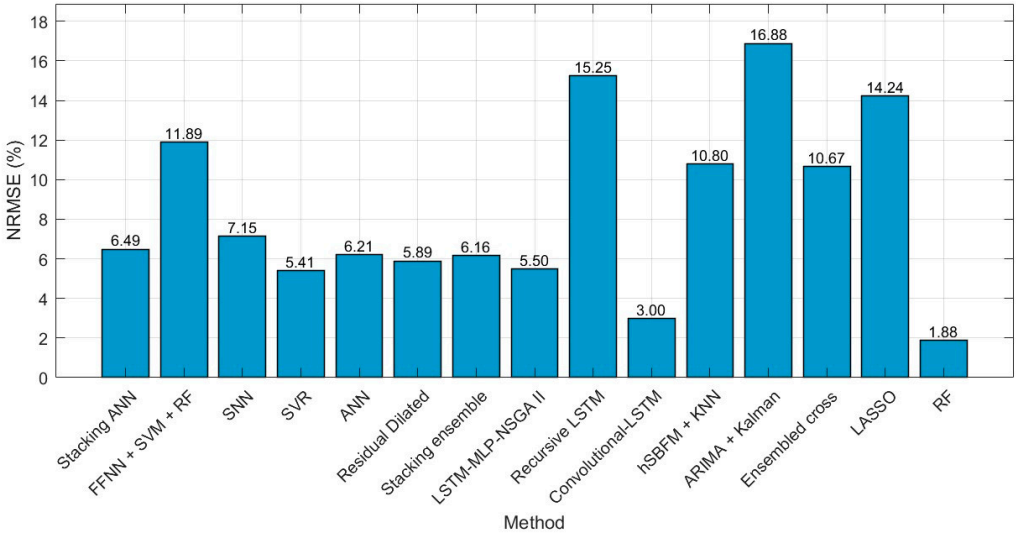


Figure 5. Distribution of the reported NRMSE (%) values.

Individual markers represent the single NRMSE value reported in each study. The median of the collected NRMSE values is 6.5%, and the interquartile range is 6.4%, with reported values spanning from 1.9% to 16.9%. This illustrates both the variability across forecasting methods and the importance of explicitly stating the chosen normalization approach in each study. Since the reported values come from different systems, datasets, and, in some cases, different normalization conventions, the figure is not a direct head-to-head benchmark. Instead, it highlights the central tendency and spread of published normalized errors and helps identify studies that report comparatively low or high NRMSEs.

The primary purpose of the "Tips" column is to highlight shading issues reported in the articles. As observed, BIPV buildings with PV modules on their façades experience a higher degree of shading, which adversely affects energy production (the majority of shading-related articles concern BIPV systems). Moreover, varying shading conditions across different façades lead to differences in their respective outputs.

The other column of the table presents case studies focused on deployment. From the extracted set, residential deployments represent the largest share (23 out of 86 studies; 26.7%), followed by research/laboratory testbeds (17 out of 86; 19.8%) and university campuses (15 out of 86; 17.4%). Industrial and mixed-community deployments are scarce (four studies each; 4.7%). This distribution indicates a dominance of small-scale and academic testbeds in the literature; large commercial and industrial deployment-focused evaluations, which are critical for grid-integration, remain underrepresented.

The final column presents the geographical focus of the studies, which was carried out in countries. Their distribution is shown in Figure 6.

In order to conduct a comparative analysis considering forecasting methods, building types, and geographical contexts, it is essential that studies employ a consistent evaluation metric. Since previous research has used a variety of metrics (in some cases, even when metrics were normalized, the units or rates were not clearly reported), R^2 emerges as the most comparable scale, ranging from 0 to 1, where 1 indicates perfect performance and 0 indicates the worst. Accordingly, studies reporting R^2 values were re-extracted and summarized in Table 9, providing a clear basis for comparison.

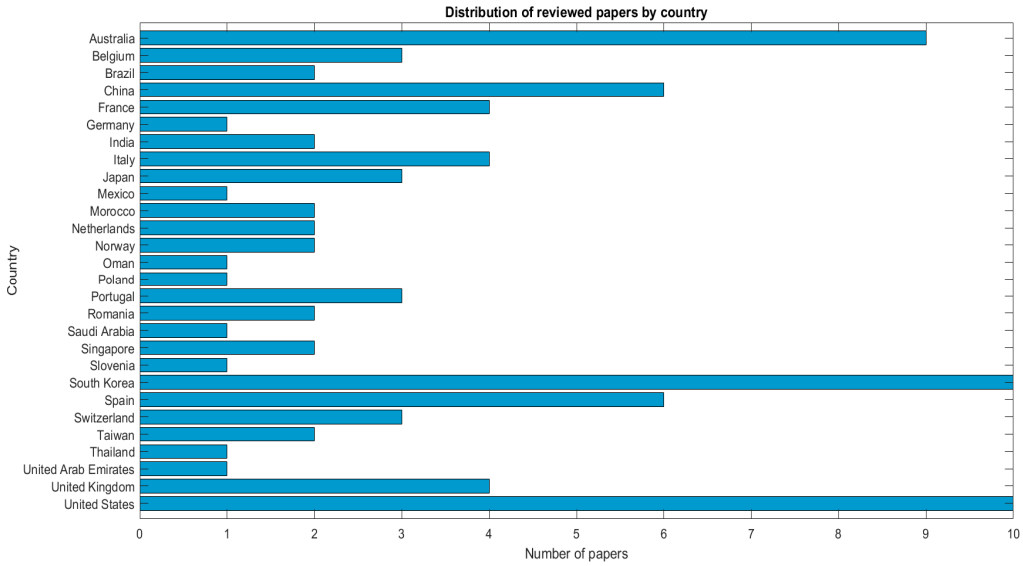


Figure 6. Geographical focus of the studies.

Table 9. Comparison of forecasting performance (R^2) across studies.

Methods	Forecasting Accuracy	Case Studies	Geographical Location	Ref.
SAPM	R^2 for each façade East = 0.73 South = 0.89 West = 0.90	University campus	Madrid/Spain	[146]
ANN	R^2 = from 0.63 to 0.88	Residential	Singapore	[147]
ANN	R^2 = 0.9768	_	Korea	[151]
NARX	R^2 = 0.95	Office	Seoul/Korea	[154]
Mathematical modeling with multiple regression	R^2 = 0.81	_	São Paulo/Brazil	[156]
ANN	R^2 = 0.8833	Residential buildings	Kovilpatti/India	[159]
CNN	R^2 = 0.96	Campus building	Strasbourg/France	[161]
Back-propagation ANN	R^2 = 0.9489	Several types	Cardiff/The UK	[169]
MLP	R^2 = 0.8481	Smart Home	Swiss	[179]
4-layer DNN	R^2 = 0.95	Retail shop	Korea	[189]
- Residual Dilated Causal Convolutional Network	R^2 = 0.9308	Residential	Tainan/Taiwan	[190]
Stacking ensemble models with RNN	R^2 = 0.86	Industrial Co.	Taiwan	[197]
LSTM + self-attentions	R^2 = 0.934	Houses	Urayasu/Japan	[200]
Singular Spectrum Analysis + BiLSTM + Bayesian Optimization	R^2 = 0.971	Real-world rooftop stations	Eastern China	[206]
ANN	R^2 = 0.935	Laboratory	Strasbourg/France	[214]
AML	R^2 = 0.997	Energy Institute	South Korea	[218]

Table 9. *Cont.*

Methods	Forecasting Accuracy	Case Studies	Geographical Location	Ref.
AML	R^2 = 0.857	University Campus	Toyama/Japan	[219]
WTEEMD + FCM + IGWO + LSTM	R^2 = 0.9915	ZEB	Alice Springs/Australia	[225]
VMD + Informer + Enhanced Chaos Game Optimization	R^2 = 0.9732	- Steel Structure Engineering Co.	Nantong/China	[226]
DSE-XGB	R^2 = 0.96	Commercial	Netherlands	[227]
SVR	R^2 = 0.934	University building	Bidart/France	[228]
LASSO	R^2 = 0.57	Industrial	Germany	[229]
RF	R^2 = 0.987	Smart buildings	Estonia	[230]
RF	R^2 = 0.99	- A single-family dwelling	Maryland/USA	[232]

Traditional statistical models, such as the SAPM and LASSO, demonstrate a variable level of performance. For instance, a study using SAPM on a university campus in Madrid reported R^2 values ranging from 0.73 to 0.90 across different building façades. However, the LASSO model in an industrial setting in Germany yielded a significantly lower R^2 of 0.57, suggesting a strong dependency on the operational environment and data characteristics.

ML models, particularly ANN, are widely used and show a large spectrum of reported accuracy. One study on residential buildings in Singapore reported R^2 values from 0.63 to 0.88, while another ANN application in Korea achieved a much higher R^2 of 0.9768. This wide performance range underscores a critical observation: the model type itself is only one of several determinants of accuracy. The performance of a model is not solely a function of its inherent sophistication but is profoundly influenced by the geographical condition, quality, and nature of the data on which it is trained and tested. Other ML methods like SVR and RF also demonstrated high performance, with RF achieving an exceptional R^2 of 0.987 in Estonia and 0.99 in the USA.

The most compelling trend from the data is the superior performance of complex, DL, and hybrid ensemble models. This category includes architectures like CNN, LSTM, and advanced hybrid models. For example, a CNN-based model on a campus building in France reported an R^2 of 0.96, while a LSTM model with self-attention in Japan achieved an R^2 of 0.934. The highest reported R^2 values are consistently associated with these more complex architectures. An AML model achieved a near-perfect R^2 of 0.997 in an energy institute setting in South Korea, and a hybrid model combining WTEEMD, FCM, IGWO, and LSTM in a ZEB in Australia reached an R^2 of 0.9915. The ability of these models to capture highly nonlinear, temporal, and chaotic patterns inherent in PV data is likely the reason for their superior performance.

The numerical superiority of DL and ensemble models, however, comes with a significant trade-off. While their performance can be exceptional, these models often function as "black boxes", making it difficult for researchers and engineers to understand the causal relationships and underlying reasons for their predictions. This lack of interpretability is a key consideration for practical implementation, where understanding model failures or the influence of specific input variables can be as important as the accuracy itself.

The findings confirm that, while complex DL and hybrid models often report the highest R^2 values, their success is not universal but is highly dependent on the opera-

tional environment. Controlled settings like laboratories and ZEBs inherently present less complex forecasting problems, allowing models to achieve near-perfect performance. In contrast, real-world scenarios such as industrial sites (owing to greater operational variability and pollution) or university campuses introduce significant variability that can limit the achievable accuracy, regardless of the model's sophistication.

Furthermore, the analysis demonstrates that geographical location serves as a critical proxy for underlying climatic conditions, which introduce a unique set of challenges to forecasting. The unique case of BIPV systems highlights the importance of considering micro-environmental conditions, even across different façades of the same building.

Finally, it should be noted that, in addition to the forecasting method, building type, and geographical context, various other factors (such as the dataset characteristics, data intervals, forecasting horizon, season, PV panel cleanliness, temperature, ventilation, etc.) significantly affect forecasting accuracy.

3.7. Real-World Use Cases of PV Forecasting in Buildings

PV generation forecasting is not limited to labs or academic research, since it also has real-world applications. An empirical study [234] evaluated PV generation forecasting on PV rooftop systems at Hong Kong University of Science and Technology using a high-resolution inverter and meteorological dataset. Three models (SVM, Random Forest, and MLP) were trained to predict normalized PV output from selected weather features. Random Forest performed best (R^2 0.96, MAE 3.6 W, RMSE 4.8 W, and MAPE 5.2%), followed by MLP and SVM. The measured annual degradation averaged 1.0%, primarily due to humidity-related potential-induced degradation, with the authors estimating that forecast-informed predictive maintenance could reduce maintenance costs by around 10–15%.

Kut et al. [235] proposed a short-term forecasting of energy production in residential PV systems, leveraging meteorological data collected in a real-world environment and applying ML techniques. Their approach could be integrated into EMS without relying on storage solutions, thereby promoting a more conscious and efficient use of local energy resources.

An important practical consequence for PV forecasting in buildings is that both system configuration (fixed vs. tracking) and the energy-balance timescale (yearly vs. monthly) materially affect forecasting needs and the types of forecast errors that are operationally relevant. D'Agostino et al. [236] showed that systems sized to achieve net zero on an annual basis can still exhibit large monthly deficits (fixed: 808.1 kWh; biaxial tracking system: 727.4 kWh when sized yearly) and that sizing to a monthly balance required substantial oversizing in their case study (fixed: 2.4 $\rightarrow$ 7.2 kW; tracking: 1.5 $\rightarrow$ 3.6 kW), with consequent increases in capital cost. Tracking PV systems can meet monthly electricity demands with fewer panels and typically outperform fixed PV systems regarding CO_2 emissions. However, solar trackers increase capital and maintenance costs and require larger spacing to avoid self-shading.

Beyond algorithmic advances, PV forecasting in the building sector must be evaluated for its role in real decision-making. D'Agostino et al. [237] applied EnergyPlus coupled with a PV model to compare multiple deployment options (roof-mounted fixed, large parking-area array, and roof-mounted tracking) alongside envelope and HVAC measures. The alternatives were evaluated using technical and economic Key Performance Indicators (KPIs), including annual electric energy savings, Net Present Value (NPV), and discounted payback period, and were ranked through a robust multiple-criteria decision-making framework that explicitly models policymaker vs. tenant preferences. The study shows how PV sizing and cost assumptions lead to divergent stakeholder rankings (a large PV

maximizes energy and CO_2 savings, while a smaller tracking rooftop system yields the best NPV for tenants), illustrating the necessity of stakeholder-aware economic metrics and robustness/sensitivity analyses when assessing PV in retrofit decisions.

Therefore, forecasting frameworks should consider forecast horizon/accuracy and match them to the intended decision (e.g., seasonal/monthly forecasts for planning and sizing; day-ahead forecasts for storage scheduling; intra-day/hour-ahead forecasts for curtailment and load-following) and be evaluated with decision-relevant KPIs and robustness/sensitivity analyses.

These studies highlight how PV generation forecasting can move beyond theoretical research to deliver tangible benefits for real-world energy management.

3.8. Emerging Time Series Forecasting Technologies

The rapid development of AI-based technologies has led to the emergence of new methods aimed at enhancing forecasting accuracy. This section reviews these advances and underscores their potential application to PV generation forecasting in the building sector.

Given widespread concerns over data protection and privacy, Google introduced FL to permit model development on large, distributed datasets in a privacy-preserving manner, avoiding the transfer of raw user data to a central server [238]. There are only a small number of published applications of FL for PV generation forecasting. Although Ref. [181] (Section 3.5.2) highlighted FL's potential to lower prediction RMSE, the limited research suggests a clear need for further evaluation, particularly for building-level deployments.

A Transformer [239] is a neural sequence transduction model that dispenses with recurrence and convolutions and instead computes representations using self-attention together with positional encodings. Attention-based Transformer models have rapidly become prominent in PV forecasting; a recent review [240] reported that Transformers often achieve the highest accuracy in PV power prediction. Refs. [199,200] (Section 3.5.2) highlighted the potential of transformer architectures and attention mechanisms to improve forecasting performance. Reflecting their growing popularity, the number of studies applying these methods to PV generation forecasting has increased in recent years.

Quantum Machine Learning (QML) is an emerging area that integrates principles of quantum computation (such as superposition, entanglement, and interference) into Machine Learning models and algorithms [241]. To the best of our knowledge, there are currently no published studies that explicitly apply QML techniques to PV generation forecasting in buildings. Nevertheless, recent works have explored QML or hybrid quantum–classical methods for closely related problems, including solar irradiance forecasting [241], building energy consumption forecasting [242], and solar power forecasting in solar farms [243]. These works provide preliminary evidence of QML's potential for time series problems.

Spiking Neural Networks, regarded as third-generation neural models, emulate biological neurons by communicating with discrete spikes [244]. Two applications include wind-speed forecasting, where Thangaraj et al. [245] reported that a Spiking Neural Network-based predictor attains a lower MSE than earlier Neural Network approaches. For short-term PV power forecasting, Chen et al. [246] combined K-Means clustering with Spiking Neural Networks and demonstrated superior accuracy relative to CNNs and SVMs.

Graph Neural Networks (GNNs) employ a message-passing scheme in which each node's representation is formed from its own attributes together with aggregated information received from adjacent nodes [247]. The graph is defined by an adjacency matrix (most studies use a fixed matrix based on statistical similarity/geographic location) [248]. Hu et al. [247] developed a data-driven model for urban building energy that integrates solar-based interdependencies into a spatiotemporal graph convolutional network. Their

experiments showed robust performance under multiple scenarios, with the model yielding 5% MAPE. Yang et al. [249] adopted a combined graph-convolutional and LSTM architecture to capture spatial shading among rooftops for spatiotemporal PV potential estimation. Their results indicated notable gains over standard LSTM models: relative error reductions of approximately 12% (MAPE), 13% (RMSE), 21% (MAE), and 22% (MSE). Although these studies demonstrate the promise of graph-based spatiotemporal models for rooftop PV estimation, the number of works that explicitly address building sector PV generation forecasting remains limited, and further research is needed.

A digital twin is a dynamic virtual model of a physical asset that enables real-time monitoring [250]. Through connectivity and fast data processing, it mirrors the building's state, maintains synchronized data, and can simulate system behavior to support operational decision-making [250]. This approach has been widely used for building energy consumption forecasting; Boukaf et al. recently published a review summarizing key advances in this area [250]. Digital-twin applications are also well established in wind energy systems [251], yet their deployment for PV systems remains limited to date [252].

Overall, these novel approaches provide exciting opportunities to enhance PV generation forecasting for buildings, though broader evaluation and practical implementation are still required.

3.9. Practical Implications of Forecasting Accuracy

The practical implications of PV generation forecasting accuracy include optimal system sizing and strategies for grid integration.

PV generation forecasting is a critical element of efficient EMS operation. El-Baz et al. [253] discussed that uncertainties in PV generation forecasts and variability lead to less efficient operation plans for household devices, thereby reducing building autonomy, self-sufficiency, and potential cost savings and self-consumption. Most studies recommend forecasting horizons of 6–48 h for effective load planning (shorter horizons may be acceptable only when power-to-heat devices such as heat pumps are absent) [253]. The typical temporal resolution for building-scale applications is 15 min–1 h [253].

Sorour et al. [254] developed an EMS for PV–battery systems and demonstrated that integrating PV forecasts into its optimization can deliver substantial economic benefits. Over six months, their proposed method reduced net energy exchange with the grid by up to 194% and total operating costs by 54% compared to the existing approaches while lowering energy bills by as much as 46%. They also showed that forecast-based scheduling is more efficient when the EMS objective explicitly minimizes the net grid exchange.

Accurate forecasting of PV generation enables the estimation of inverter capacity and the optimal sizing of the overall energy system. Despite its importance, there is a striking lack of studies addressing forecast-driven system sizing in the building sector. Reliable predictions could help designers balance investment costs with operational performance, avoiding undersized installations that fail to meet demand or oversized systems that increase unnecessary expenses. Moreover, precise forecasts support integration with building energy management systems, enabling efficient load scheduling and storage utilization to maximize self-consumption and reduce grid dependence. Given the growing adoption of rooftop PV systems in buildings, research on forecast-informed sizing strategies is essential to achieve cost-effective, resilient, and sustainable energy solutions.

Forecast accuracy directly influences how effectively renewable energy can be integrated into the grid [255]. In systems with high wind shares, integration is highly sensitive to forecast errors, while PV integration is less dependent on accuracy. As renewable penetration rises, forecast errors increase the demand for control reserves, though this effect stabilizes once the installed capacity reaches the annual peak demand. Furthermore, the

length of the forecast horizon is critical, since better-aligned forecasts improve predictive dispatch and overall system stability.

4. Discussion

The discussion section seeks to present the principal findings related to this topic. Through a summary and analysis of the existing literature in Section 3, researchers can pinpoint the most effective methodologies and strategies for PV generation forecasting while also identifying significant challenges and opportunities for future exploration. This section presents the main outcomes of the study, and the identified research gaps are evaluated.

4.1. Outcomes

This study explores key aspects such as the significance of high-quality data in PV generation forecasting, various forecasting methodologies, evaluation approaches (metrics and post-processing), challenges in modeling PV systems in the building sector, and a comprehensive review of strategies related to BIPV and BAPV systems. Building on the previous sections and the conducted analysis, the following outcomes were verified.

Factors influencing forecasting accuracy include the number of variables, as an increase in variables does not necessarily lead to a relative improvement in prediction accuracy. While using larger sets of input variables resulted in the smallest forecast errors, the computational time increased as the number of input variables grew. Furthermore, improving prediction accuracy is not necessarily related to the complexity of the model, and in some cases, simpler models may yield better results. In this regard, the dataset plays a key role.

Factors influencing the impact of environmental and system characteristics include the azimuth angles and tilt in PV systems, which influence the energy yield and power profiles and play a crucial role in the design and accuracy of PV output forecasting models. In addition, the accuracy of solar radiation forecasting will directly influence the framework's forecasting of PV output capability. PV generation is more consistent on sunny days, leading to better prediction performance compared to cloudy days. Despite the expectation of more sunshine in the summer, some locations experience their wettest season during this time, leading to reduced sunshine. Consequently, summer weather patterns in these areas may resemble those of cold months. Therefore, the location is crucial, as seasonal changes may not always correlate with weather conditions.

The selection of models is influenced by the balance between accuracy and computational time. While the hybrid model is acknowledged for its superior accuracy in most cases, it compromises speed compared to individual models. On the other hand, when forecasting methods are applied to building applications, some ML methods show superior performance, suggesting that more complex approaches do not always yield the best accuracy. Enhancing the accuracy of PV output prediction is achievable by exploiting the temporal correlations within the PV generation data and spatial similarities. Another challenge is that the accuracy of NWP-independent models deteriorates after a few hours, resulting in lower reliability compared to NWP-dependent models.

The prediction models for BAPV and BIPV systems include physical, statistical, hybrid, and ensemble models. Based on Tables 5 and 6, it was possible to conclude that physical models are rarely used due to their reliance on detailed system parameters, which are often unavailable or difficult to measure. Statistical models, such as LSTM, are widely applied due to their ability to capture temporal dependencies and handle time series data effectively. Other Neural Networks models, like ANN and MLP, are popular for modeling nonlinear relationships, while SVR and SVM are frequently used as benchmarks for their robustness and reliable performance. Hybrid models, like CNN-LSTM, effectively integrate

spatial and temporal features, addressing the complexity of PV forecasting tasks. Other hybrid approaches, such as LSTM-based combinations with optimization techniques, offer improvements in accuracy but increase the computational time. Ensemble methods, like RFs and XGBoost, are prominent for combining weak learners and enhancing accuracy.

In terms of performance evaluation, RMSE is the most widely used metric in PV generation forecasting due to its ability to penalize larger errors, making it critical for energy management. Its interpretability, being in the same units as the predicted values, also adds to its popularity.

4.2. Gaps and Future Work

Although extensive research has been conducted on the forecasting of PV generation in buildings, certain gaps persist, requiring further exploration. This section outlines key areas where additional investigation could yield valuable insights into improving the accuracy, efficiency, and applicability of PV generation forecasting methodologies in the building sector.

Priority roadmap. Among the identified gaps, immediate priority should be given to (1) translating PV forecasts into operational EMS decisions (to quantify peak shaving, cost, and sizing benefits); (2) addressing methodological weaknesses through standardized benchmarks and rigorous comparative studies (including missing data handling and spatial–temporal modeling); and (3) advancing algorithmic capabilities (online/federated learning and robust feature selection). Subsequent priorities include targeted studies on façade temperature mismatch, soiling and equipment aging, systematic investigations of localized environmental factors, and the development of BIPV/bifacial-specific forecasting datasets.

Although the majority of reviewed studies focused on developing and fine-tuning forecasting models for PV generation, there was a clear and pressing gap in translating those forecasts into operational decisions within BEMSs. Future research should treat PV forecasts as inputs (and not as final results) and rigorously evaluate how forecast-driven EMS strategies affect peak shaving, electricity bill reduction, operational costs, and technoeconomic outcomes such as energy storage deployment and PV/inverter sizing.

The methodological gaps include several critical issues, such as the limited comparison of forecasting methodologies, the assumption of constant PV generation despite real-world variations, the neglect of spatial and temporal correlations, the lack of missing value imputation strategies, and the underutilization of heuristic optimization techniques. Many studies have conducted comparisons among a restricted set of methodologies, indicating the need for researchers to undertake comprehensive assessments to identify the most reliable and efficient models, thereby advancing forecasting methodologies. Additionally, most forecasting models assume constant PV generation, disregarding real-world variations (ignoring panel degradation, inverter losses, and temperature-dependent performance variations) and leading to inaccurate forecasts. Furthermore, the focus of many PV output forecasting methods on specific PV systems neglects spatial and temporal correlations among numerous systems, which is essential for improving forecasting accuracy. Moreover, the scarcity of literature discussing missing value imputation in building PV output forecasting further obstructs the development of robust methods for handling missing data in real-world datasets. Addressing this gap is essential to improving the accuracy of PV generation forecasts. Lastly, heuristic optimization techniques, despite their proven potential to improve the accuracy of forecasting models, are rarely applied in the literature. These methods could offer significant benefits in optimizing model parameters and enhancing predictive performance, particularly for complex PV systems.

Advancing methods in PV forecasting involve addressing several key challenges, such as the reliance on traditional offline ML, the need for online learning techniques, the lack of backward feature elimination, and the limited availability of standardized data. Additionally, privacy concerns around data sharing could be addressed using federated learning algorithms to improve forecasting accuracy. The common use of traditional offline ML in PV generation forecasting limits the effectiveness of these models in real-world applications. Further research on online learning techniques for ML in PV generation forecasting can improve the accuracy and applicability of these models. Online models, which regularly update with both predicted and real values, could play a significant role in future literature. In addition, the lack of application of backward feature elimination techniques hinders the identification of the most relevant set of features, potentially limiting the accuracy and efficiency of the predictive model. Finally, the limited availability of comprehensive and standardized data on building PV output patterns restricts the development and deployment of reliable models for PV output forecasting. Despite building owners having the data, they are hesitant to share them due to privacy concerns. Future research could investigate federated learning algorithms and their potential to improve energy prediction accuracy while preserving privacy.

Due to varying ventilation conditions, significant temperature differences, and uneven temperature distribution across PV façade modules, substantial mismatch power losses occur, adversely impacting electrical performance. Future research should focus on analyzing the temperature distribution of PV façades to mitigate these losses. Additionally, PV system efficiency declines over time due to equipment aging and the accumulation of surface dust. Therefore, future evaluation methods should consider the system's full life cycle and its interaction with the BEMS.

Environmental and operational factors affecting PV generation include factors such as partial shading, building orientation and tilt, building height, proximity to trees and other obstructions, dust and dirt, snow and ice, and pollution. However, a noticeable gap remains in comprehensive investigations into the impact of these localized factors. This lack of research limits our understanding of their effects on PV generation in building environments.

The insufficient research on BIPV, despite the growing importance of integrating solar technology into building structures, hampers the development of innovative BIPV solutions and the advancement of the intersection of PVs and building technology. Moreover, the significant gap in research specifically investigating the energy forecasting of bifacial solar PV systems in buildings obstructs the development of accurate forecasting models. However, these critical aspects have been overlooked in the current literature.

5. Conclusions

To tackle the growing challenges of energy consumption and GHG emissions, buildings play a crucial role in transforming the energy sector by facilitating the integration of PV systems. By accurately predicting PV generation alongside building energy demands, energy management strategies can be optimized. This paper highlights the significance of forecasting PV generation in buildings and underscores the critical need for reliable data. It also clarifies the key distinctions between PV forecasting in the building sector and other sectors. Moreover, various forecasting methodologies and evaluation metrics are thoroughly examined.

A comprehensive review of the current landscape of PV generation forecasting in buildings was conducted. The reviewed articles were categorized into BIPV and BAPV using a content analysis approach. This review examined a range of forecasting methods, including physical models and statistical approaches (such as Artificial Neural Networks (ANNs) and Machine Learning (ML)), as well as hybrid and ensemble techniques, within

both BIPV and BAPV contexts. Statistical methodologies dominate due to their adaptability and ability to capture nonlinear relationships. While physical models offer deterministic accuracy, their reliance on detailed input parameters limits practical applications. Hybrid and ensemble models, despite their enhanced predictive capabilities, remain computationally demanding. Additionally, this study analyzed the most frequently used evaluation metrics. RMSE emerged as the most widely used evaluation metric, reflecting its effectiveness in penalizing large forecasting errors, though other metrics, like MAE, are also employed for complementary insights.

Recognizing the vast potential of PV generation forecasting, this article presents in-depth discussions on existing research gaps (such as the impact of localized environmental factors, the need for advanced forecasting methodologies, and data privacy challenges) and outlines key future research directions (including the development of online learning models, improved missing value imputation techniques, and applying heuristic optimization) for energy researchers to explore.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ANN	Artificial Neural Networks
ARIMA	Autoregressive Integrated Moving Average
AML	Automatic Machine Learning
BiLSTM	Bidirectional Long Short-Term Memory
BAPV	Building Attached Photovoltaics
BEMS	Building Energy Management Systems
BIPV	Building Integrated Photovoltaics
CSO	Competitive Swarm Optimizer
CHAID	CHi-square Automatic Interaction Detection
CART	Classification And Regression Trees
CEEMDAN	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise
CNN	Convolutional Neural Network
ConvLSTM	Convolutional Long Short-Term Memory

DTR	Decision Tree Regression
DTs	Decision Trees
DL	Deep Learning
DNN	Deep Neural Networks
DR	Demand Response programs
DSE-XGB	Improved generally applicable stacked ensemble algorithm
ESS	Energy Storage Systems
ETs	Extra-Trees
FL	Federated Learning
FFNN	Feed-Forward Neural Network
FCM	Fuzzy C-Means
GA	Genetic Algorithm
GNNs	Graph Neural Networks
GRU	Gated Recurrent Unit
GWO	Grey Wolf Optimizer
HVAC	Heating, Ventilation, and Air Conditioning
hSBFM	hierarchical Similarity-Based Forecasting Models
IGWO	Improved Gray Wolf Optimization
KNNs	K-Nearest Neighbors
KPIs	Key Performance Indicators
LASSO	Least Absolute Shrinkage and Selection Operator
LR	Linear Regression
LSTM	Long Short-Term Memory
ML	Machine Learning
MLPE	Module-Level Power Electronics
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MBE	Mean Bias Error
MLP	Multi-Layer Perceptron
MVR	Multi-Variable Regression
MLR	Multiple Linear Regression
NARX	Nonlinear Autoregressive with exogenous input
NPV	Net Present Value
NRMSE	Normalized Root Mean Squared Error
NWP	Numerical Weather Prediction
PV	Photovoltaic
QSVM	Quadratic Support Vector Machine
QR	Quantile Regression
QRF	Quantile Regression Forests
QML	Quantum Machine Learning
RBF-ANN	Radial Basis Function Artificial Neural Networks
RFs	Random Forests
RNN	Recurrent Neural Networks
RMSE	Root Mean Squared Error
SAPM	Sandia Array Performance Model
SBFMs	Similarity-Based Forecasting Models
SC*	Sky Condition
SVF	Sky View Factor
SNN	Standard Neural Network
SVM	Support Vector Machine
SVR	Support Vector Regression
R^2	The Coefficient of Determination

TBATS	Trigonometric Seasonal, Box-Cox Transformation, Autoregressive Integrated Moving Average residuals, Trend, and Seasonality
VMD	Variational Mode Decomposition
WTEEMD	Wavelet Threshold Improvement Ensemble Empirical Mode Decomposition
XGBoost	Extreme Gradient Boost

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